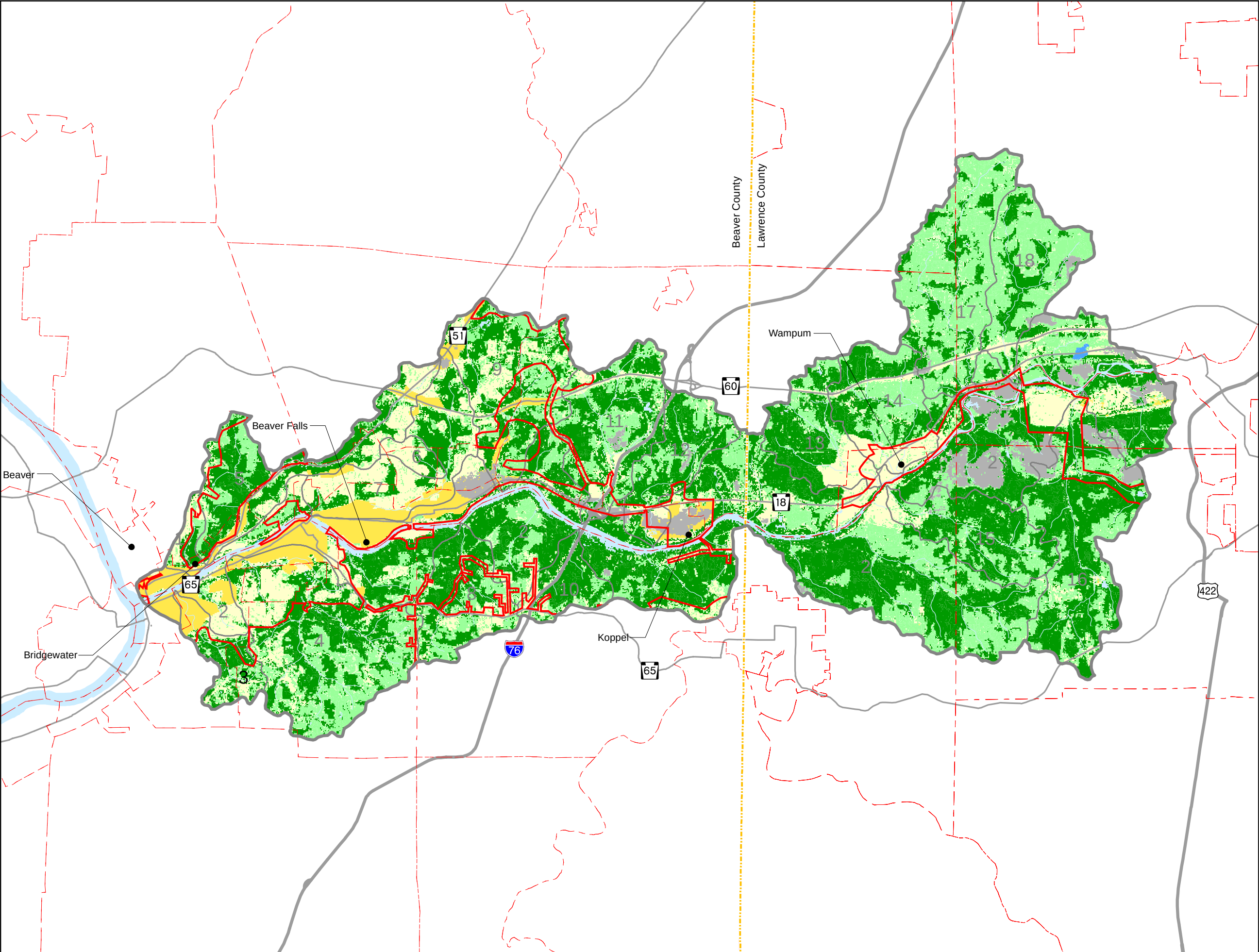




Legend

Study Area Boundary

Watershed Boundary

County Boundary

Municipal Boundary

Rivers/Streams/Lakes

Major Highways

Major Roads

Sanitary Sewer/Water Service

Land Coverage

Water

892 ac.

Emergent and Woody Wetlands

45 ac.

Coal Mines and Quarries

2,175 ac.

Row Crops and Hay Pastures

16,428 ac.

Deciduous, Coniferous and Mixed Forests

25,310 ac.

Low Density Urban

8,164 ac.

High Density Urban

2,760 ac.

Transitional

2,535 ac.

Note:
Land coverage information was provided by Penn State University.

Sub-Watersheds

1. Hamilton Run

2. Beaver River

3. McKinley Run

4. Block House Run

5. Brady Run

6. Grimms Run

7. Walnut Bottom Run

8. Bennett Run

9. Wallace Run

10. Thompson Run

11. Clarks Run

12. Stockman Run

13. Wampum Run

14. Eckles Run

15. Snake Run

16. McKee Run

17. Jenkins Run

18. Edwards Run

LAND COVERAGE INVENTORY

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council

Prepared by: Environmental Planning and Design, LLC

Date: September 2006

1989.06.02r2

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NORTH

Existing Development Tools and Land Use

Development Tools

In Pennsylvania, most decisions concerning land use are ceded to local governments. There are many tools available to local governments to help plan or manage their growth or to protect and maintain a quality of life.

Primary among them is the Pennsylvania Municipalities Planning Code (MPC), which outlines the controls that may be placed on private lands and how those controls are to be carried out. Four of the most prominent tools are: the Comprehensive Plans, Zoning, Subdivision and Land Development Ordinances, and Planning Commission. While these land use tools may be controversial in parts of Pennsylvania, they can protect a community's character and quality of life.

1. Comprehensive Plan

A Comprehensive Plan is an important land use development tool as it “serves as a policy guide to decision making about physical development in the community. It is an explicit statement of future goals for the community and serves as a formal vision for the planning commission, elected officials, and other public agencies, private organizations, and individuals. A community's comprehensive plan provides context and direction for a community's land use ordinances and regulations and should be updated and modified continuously in response to changes in the community.”⁴

2. Zoning

Zoning regulates land use using designated districts designed to segregate incompatible uses. Municipal zoning is employed to protect a municipality's safety health and welfare as it relates to current and future land development.

3. Subdivision and Land Development Ordinance

A Subdivision and Land Development Ordinance outlines development guidelines and provisions in more detail and may provide for specific mechanisms for controlling site development patterns and design standards.

4. Planning Commission

Planning commissions “are advisors to their elected governing body on matters concerning the physical development of the community.” This includes land use regulations, building structures, and developing recreation plans.

In addition to the tools outlined above, Act 148 of 1973 authorizes municipalities to form Environmental Advisory Councils (EACs). An EAC is a board of qualified individuals appointed to oversee and give advice on environmental conservation and improvement in a certain area. Many successful EACs in Pennsylvania help their respective communities with a wide range of environmental activities.⁵

⁴ Definition from the document *An Inventory of Planning in Pennsylvania, Pennsylvania*, published by the State College of Agricultural Sciences, 2001.

⁵ See the EAC Network website: www.greentreks.org/eacnetwork

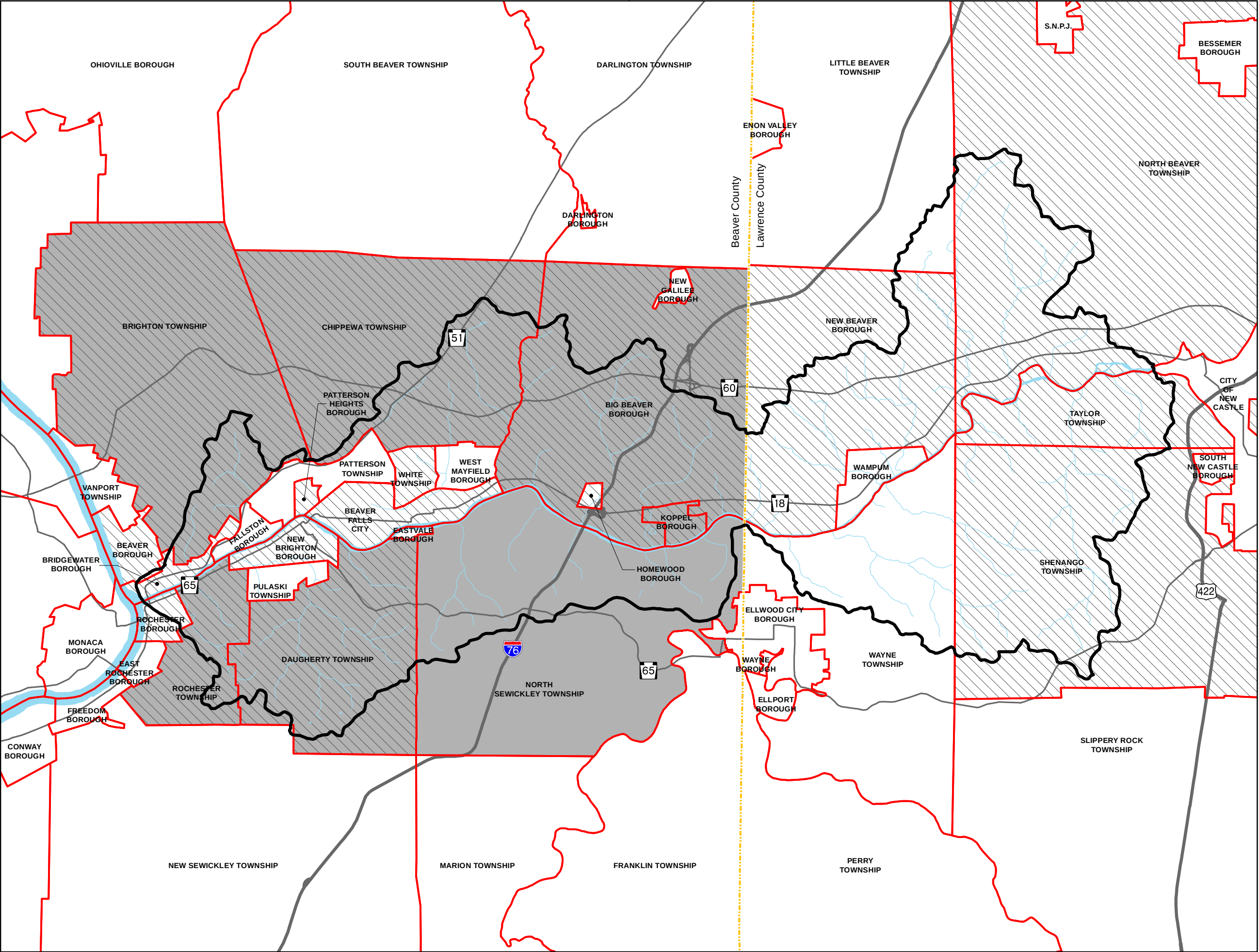
Communities may receive technical and financial assistance to implement these tools through programs offered by the Pennsylvania Department of Economic Development (www.newpa.com) or the Local Government Academy (www.localgovernmentacademy.org).

Table 4-3 and the Municipal Land Use Regulations Map identify the land use regulations that are currently in place within the Study Area's municipalities.

Table 4.3: Municipal Ordinance Elements Analysis

		Methods for Designating Sensitive Areas						Sample Design/Review Requirements to control impacts of development in sensitive areas					Notes
Municipality	Ordinance Type	Floodplain District/Section within the Ordinance	Floodplain Section is a Zoning Overlay	Special Environmental Zoning District	Boundaries determined by the 100-yr flood from Flood Insurance Study &/or Flood Hazard Boundary Map by FEMA)	Boundaries also determined by State or Federal definitions of Wetlands	Other General Criteria	Site plan review	Setback requirements from waterways, floodplains/floodways, etc.	Maximum building coverage requirements	Maximum impervious surface requirements	Additional land use requirements	
Beaver Borough	Zoning	X			X							X	General Floodplain, Floodway, and Fringe Districts
Beaver Falls City	Zoning												Mentioned under community development objectives section, but no formal regulations
Big Beaver Borough	Subdivision				X		X						Includes provisions for flood prone and mud-slide prone areas
Big Beaver Borough	Zoning		X		X			X	X	X		X	No development within 50 of top bank without permit
Bridgewater Borough	Zoning		X		X	X			X	X		X	
Brighton Township	Zoning	X		X	X		X			X		X	Indicates a “special flood hazard area” on the zoning map; Twp has a Flood Management ordinance
Brighton Township	Subdivision	X											

Chippewa Township	Zoning	X			X		X			X		X	
Chippewa Township	Subdivision												Required to designate floodplain as part of Stormwater Management Plan
Daugherty Township	Zoning									X			
Koppel Borough	Zoning									X			
New Beaver Twp	Floodplain				X								
North Beaver Twp	Subdivision						X						Environmental Protection section addresses grading
North Sewickley Township	Subdivision				X								
Patterson Heights Borough	Zoning									X			
Patterson Township	Zoning									X			
Rochester Township	Subdivision						X						Flooding addressed in purpose statement and definitions
White Township	Zoning									X			



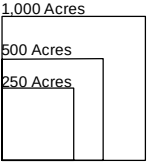
Legend

- Study Area Boundary
- County Boundary
- Municipal Boundary
- Rivers/Streams/Lakes
- Major Highways
- Major Roads

Land Use Regulation

- Municipal Zoning Ordinance
- Municipal Subdivision and Land Development Ordinance

Land use regulation information for municipalities within the study area is based on the Commonwealth of Pennsylvania's e-library (<http://www.elibrary.state.pa.us/elibpub.asp>).



MUNICIPAL LAND USE REGULATIONS
BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN
 Prepared for: Pennsylvania Environmental Council
 Prepared by: Environmental Planning and Design, LLC

Date: September 2006
 1989.06.30

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 NORTH

Act 2 and the Land Recycling Program

Act 2, or the Land Recycling and Environmental Remediation Standards Act, was signed into law by Governor Tom Ridge in 1995. This act established the Land Recycling Program, which encourages the voluntary reuse of contaminated industrial lands. Encouraging the development of these lands, commonly referred to as brownfields, offers many benefits including cost-efficient development due to the existing infrastructure on the land and the preservation of farmland, forested areas, and open space from development.

Act 2 offers incentives for adaptive reuse of contaminated sites – they are:

- Uniform cleanup standards
- Liability relief
- Standardized reviews
- Financial assistance

Sites Completed:

Big Beaver – Special Metals OPR/Rt. 18, Koppel and Westgate Business Park
Rochester – Howard E. Stuber

Sites In Progress:

Beaver Falls – Big Beaver Falls Area School District
Beaver Falls – Republic Technologies International Facility

Industrial Sites for Sale

There are several sources of information on land for sale.

- The Port of Pittsburgh lists on its website (www.port.pittsburgh.pa.us) riverfront properties for sale.
- The Pennsylvania Site Finder is a statewide program that aims to recycle properties instead of developing on greenfields (undeveloped open space). The inventory of sites is at www.pasitefinder.state.pa.us.
- The Beaver County Corporation for Economic Development lists sites on their website (www.beavercountyced.org) and maintains a database of additional commercial/industrial properties.
- The Pittsburgh Regional Alliance has a property searchable website (www.pittsburghprospector.com) with an interactive Internet mapping program.

Waste Management***Landfills and Recycling***

Title 25, Chapter 271 of the PA Code lays out the general provisions for municipal waste management. Waste disposal and landfill activities must obtain the proper permits and work through the DEP. There are no active landfills in the study area. However, there is a proposed municipal waste transfer facility for 60 acres in the Boroughs of Big Beaver and West Mayfield and the Township of Chippewa. This facility will be used to consolidate waste from the local area and will accept and transfer recyclable materials to a material recovery facility for processing.

In an effort to reduce the waste stream and to promote recycling, Act 101 – the Municipal Waste Planning, Recycling, and Waste Reduction Act – was established in 1988. Act 101 mandates recycling in Pennsylvania's larger municipalities, requires counties to develop municipal waste management plans, and provides for grants to offset expenses.

The goals of the Act are to reduce Pennsylvania's municipal waste generation; recycle at least 25 percent of waste generated; procure and use recycled and recyclable materials in state governmental agencies; and educate the public as to the benefits of recycling and waste reduction. (The benefits of recycling and waste reduction include reduced pollution risks; conservation of natural resources, energy, and landfill space; and reduced disposal costs.)⁶

The counties in this Study Area each have a recycling program. Beaver County Department of Waste Management operates a recycling center and an old newsprint animal bedding program at Brady's Run Park. See www.co.beaver.pa.us/wastemanagement for more information. Lawrence County offers a Big Blue Bin Recycling Program. See www.co.lawrence.pa.us/Recycling for information about their program.

Illegal Dump Sites

Illegal dump sites pose numerous threats, including soil and water contamination. They threaten human and animal health and lower property values. The clean up of illegal dumpsites can be very costly. The following organizations are addressing the problem of illegal dumping areas:

- PA Cleanways is a non-profit organization whose mission is to empower people to eliminate illegal dumping and littering in Pennsylvania. Beaver and Lawrence Counties have a combined chapter, which recently completed a Lawrence County Illegal Dump Survey (2008). Several dump sites have been identified within the study corridor, see Appendix 1. Every county should be surveyed by 2012. See www.pacleanways.org for a copy of the Lawrence County Illegal Dump Survey.
- PA DEP sponsors an annual River Sweep to clean up debris along rivers and streams in six southwestern Pennsylvania counties. In 2004, approximately 600 volunteers removed more than 60 tons of trash along the rivers in Allegheny, Armstrong, Beaver, Greene, Washington, and Westmoreland Counties. See www.dep.state.pa.us for more information.
- PA Resources Council is a non-profit organization that works to reduce litter and promotes recycling throughout the state. They sponsor hard-to-dispose-items collection events for large appliances, rimless tires, latex paint, electronics, and cell phones. See www.prc.org for more information.
- Construction Junction is a non-profit business that buys and sells construction debris and materials, such as cabinets, doors, windows, and lumber. See www.constructionjunction.org for more details.

⁶ <http://www.dep.state.pa.us/dep/deputate/airwaste/wm/RECYCLE/FACTS/Act101.htm>

Hazardous Waste

Hazardous waste sites and landfills near rivers and tributaries have the potential to contaminate surface and underground water supplies via runoff and leaching through the soil into the water table or aquifer. Aside from these potential threats to surface and ground water, landfills are also a possible danger to those who live, work, or play nearby. Therefore, the U.S. Environmental Protection Agency (EPA) has promulgated the following laws to deal with hazardous waste problems.

The Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) (also known as Superfund) is a national program administered by the EPA to clean up hazardous waste sites that were contaminated before 1980. These sites are commonly abandoned industrial lands or landfills where disposal of hazardous material occurred prior to existing laws that regulated industrial activities and disposals. The National Priority List (NPL) contains those sites that are being remediated first due to the severity of their status. Table 4-4 shows the sites within this study area. None are on the NPL list.

Table 4-4 CERCLA Sites in the Beaver Corridor			
Site Name	Location	EPA ID#	NPL
Beaver Sand Company Quarry	Rte 68 Beaver, Pa 15009	<u>PAD987269982</u>	No
Hydril Co	Rochester Rd Twp Rd 04122 Rochester, Pa 15074	PAD004340279	No
Vermiculite Jp1	J.P. Austin Beaver Falls, Pa 15010	PAN000305595	No
Source: EPA Envirofacts Warehouse < www.epa.gov >			

The Commonwealth of Pennsylvania, under its Land Recycling and Environmental Remediation Standards Act, and the Hazardous Sites Cleanup Act, has the authority to order clean-ups of hazardous sites that are not normally included under CERCLA. The sites in Table 4-5 have been remediated at a cost of less than \$2 million.

**Table 4-5
Hazardous Sites Response Action List**

County	Facility Name	Municipality	Response Action Name	Start Date	Finish Date
Beaver	Big Beaver Falls Area Sch Dist Lrcl	Beaver Falls	Investigation	8-Dec-00	1-Jun-03
Beaver	Ing Rich	Beaver Falls	Waste/Soil Cleanup	8-Apr-94	19-Sep-94
Beaver	Marino Bros Scrap Yard	Rochester	Investigation	6-Nov-00	15-Feb-01
Beaver	Marino Bros Scrap Yard	Rochester	Soil Removal For Trench	31-May-02	3-Jul-02

Source: EPA Envirofacts Warehouse <www.epa.gov>

The Federal Resource Conservation and Recovery Act (RCRA) requires the permitting of all hazardous waste handlers, including generators, transporters, treaters, storers, and disposers. States may administer their own RCRA permitting as does Pennsylvania, but still must report to EPA. There are too numerous RCRA-permitted facilities in this corridor to list in this Conservation Plan. To view those facilities, visit www.epa.gov/enviro/index_java.html

Toxics Release Inventory⁷

Following a fatal chemical-release accident in Bhopal, India, the Emergency Planning and Community Right-to-Know Act (EPCRA) was enacted in 1986 to promote emergency planning, to minimize the effects of an accident such as occurred at Bhopal, and to provide the public with information on releases of toxic chemicals in their communities.

Section 313 of EPCRA established the Toxics Release Inventory (TRI), which is a database on releases and transfers of more than 600 toxic chemicals generated by industrial facilities. The Pollution Prevention Act of 1990 requires that additional information on waste management and source reduction be included in the TRI. This information is gathered yearly.

Users of TRI information should be aware that TRI data do not reveal whether or to what degree the public is exposed to listed chemicals. TRI data, in conjunction with other information, can be used as a starting point in evaluating exposures and risks. The determination of potential risk to human health and/or the environment depends upon many factors, including the toxicity of the chemical, the fate of the chemical in the environment, and the amount and duration of human or other exposure to the chemical.

For more detailed information on the TRI, visit: <http://www.epa.gov/tri>

⁷ Summary taken from www.epa.gov/tri/

Critical Areas

Flood Prone Areas

Floodplains are the low-lying lands along a stream or river that are most prone to flooding. Building in floodplains is common, but often leads to heavier flooding due to 1) the loss of riparian (streamside) vegetation, which normally helps to absorb excess waters, and 2) the cumulative effect of runoff from impervious surfaces, such as houses, streets, driveways, and parking lots throughout the watershed.

Under the U.S. Army Corps of Engineers (ACE), Section 205 of the Continuing Authorities Program allows for funding for small local flood damage reduction projects. The money allows for the study, design, and construction of projects that protect communities from flooding. In addition, the National Flood Insurance Program provides coverage for flood victims; however, insurance is only provided if local communities enact and enforce land-use controls in flood-prone areas.⁸ The Pennsylvania Flood Plain Management Act (Act 166) requires municipalities in identified flood plain areas to adopt floodplain management ordinances, codes, or regulations.

The ACE also maintains a website with timely information on the conditions of reservoirs, rivers, and streams, as well as a release forecast for its reservoirs. Daily information may be obtained from www.lrp.usace.army.mil/.

Landslides

Areas of western Pennsylvania are highly susceptible to landslides due to a combination of a humid temperate climate, locally steep and rugged topography, weak rock strata, springs, and a great diversity in the weathering and erosion characteristics of near surface sedimentary rocks. In addition, landslides can be triggered by

- Addition of fill, which increases the stress on underlying materials,
- Changes in quantity or direction of water flow,
- Surface and subsurface excavations (including coal removal),
- 'Red Beds' – bedrock in hillsides composed of claystones and shales that are 40-60 feet deep. This bedrock weathers easily, especially when wet, and causes unstable slopes. Stabilization and repair can cost thousands to millions of dollars.

Because steep slopes are more susceptible to landslides, they are often not developed; therefore, they are generally better suited for woodland and wildlife habitat.

Natural Heritage Areas

These are areas that are important due to the presence of high biological diversity, a rare or exemplary natural community, a species of special concern, or for a particular use, such as nature study or instruction. More information about these areas can be found later in this Chapter.

Abandoned Mine Lands

Southwestern Pennsylvania's long history of coal mining and other mineral extraction has left a legacy of abandoned mines. The Abandoned Mine Land Inventory System in the Office of

⁸ Flooding in Western PA. Pittsburgh Geological Society. www.pittsburghgeologicalsociety.org.

Surface Mining compiles lists of problem areas. The tables in Appendix 2 list the mine problem areas in Beaver and Lawrence Counties.

Water Resources

Water Quality

Clean Water Act

The Federal Clean Water Act (CWA), which is carried out by the PA Department of Environmental Protection (DEP) under the Clean Streams Law, provides regulations that strive to “restore and maintain the chemical, physical, and biological integrity of the nation’s waters.”⁹ Regulations dealing with water quality standards of the rivers and streams in this Study Area are found in The Pennsylvania Code Title 25, Chapter 93. <http://www.pacode.com/>

All surface waters in Pennsylvania have been assigned *statewide water uses*. All surface waters should be able to support these *uses*: aquatic life, water supply, and recreation. In addition to meeting the standards for each of these statewide uses, some water bodies meet standards that make them eligible for other uses, or *designations*. The Beaver River and its tributaries and their designations are listed in Table 4-6.

Table 4-6
Tributaries and Chapter 93 Designated Uses

HQ = High Quality WWF = Warm Water Fishery TSF = Trout Stocking Fishery
CWF = cold water fishery N = Navigation

Tributary Name	Designated Use
Beaver River	WWF, N
McKee Run	WWF
Edwards Run	WWF
Jenkins Run	WWF
Snake Run	WWF
Eckles Run	WWF
Wampum Run	WWF
Stockman Run	WWF
Clarks Run	WWF
Thompson Run	WWF
Wallace Run	WWF
Bennett Run	WWF
Grimms Run	WWF
Walnut Bottom Run	WWF
Blockhouse Run	WWF

⁹ Section 101 (a)(2) Clean Water Act

Brady Run	TSF
Hamilton Run	WWF
McKinley Run	WWF
Source: http://www.pacode.com/secure/data/025/chapter93/chap93toc.html	

Sources and Types of Water Pollution

Pollution entering our waterways is typically assigned to one of two categories: point or non-point source pollution. Point source pollution comes from a defined point, such as a pipe, along a waterway. Permitted point source discharges from industrial, commercial, and municipal facilities are described below. Conversely, non-point source pollution comes from non-specific areas such as agricultural runoff and parking lots and is therefore more difficult to control and regulate.

The following sections describe both pollution sources in more depth.

Point Sources

In order to control and regulate the amount and types of pollution entering our waterways, and to help achieve designated uses and prevent water quality degradation, point sources of pollution must have proper permits to discharge wastes into the nation's waters. The National Pollutant Discharge Elimination System (NPDES) is a permitting system that targets point source dischargers, such as industrial facilities and wastewater treatment plants. Permitted facilities must meet stringent effluent limits and are responsible for monitoring water quality and reporting to the DEP. These permits are referred to as "individual" permits. For other point discharges, such as stormwater pollution or construction site runoff, a general permit is issued. General permits usually apply to smaller operations and are less stringent in the monitoring and reporting requirements.

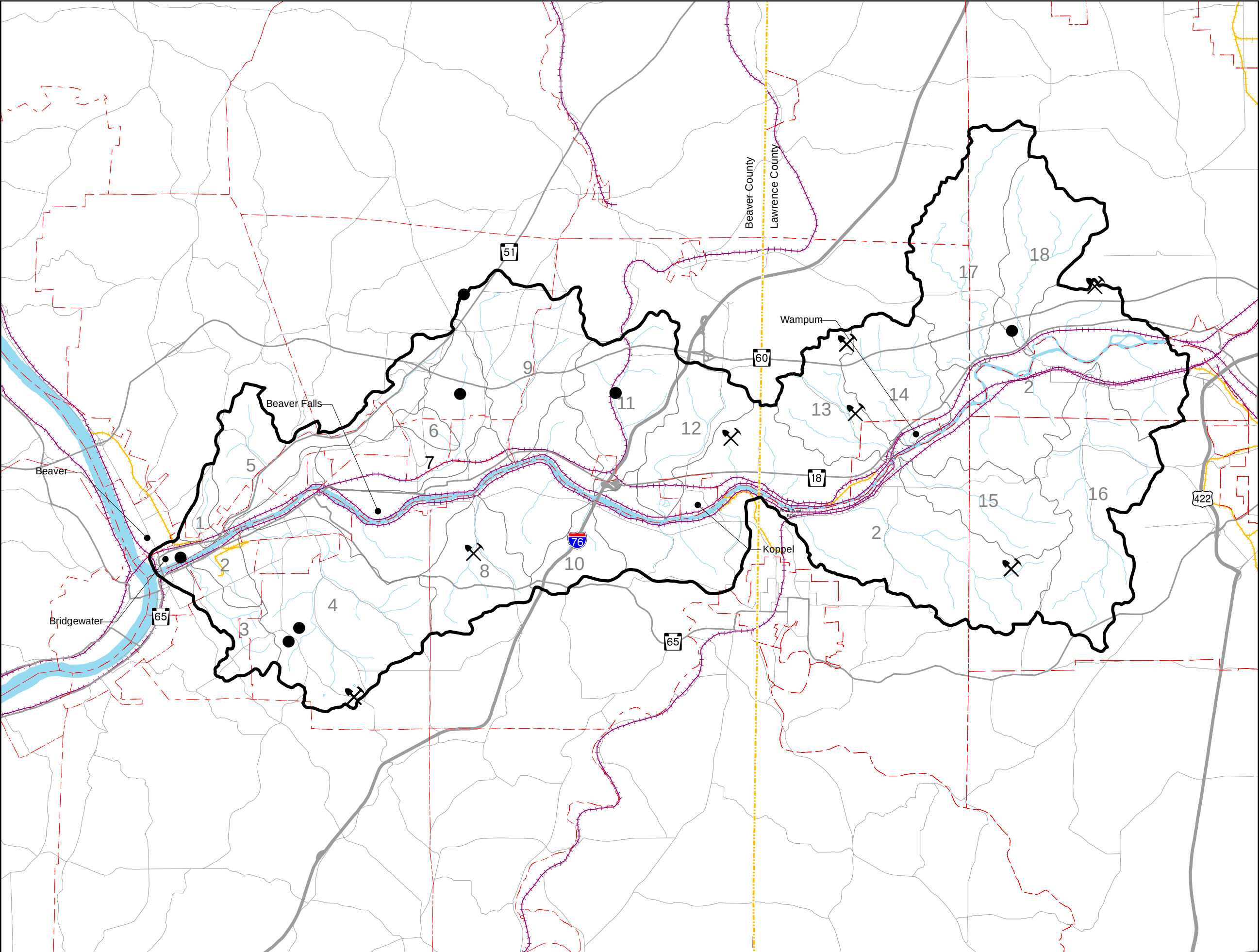
The DEP eFACTS (environment, facility, application, compliance tracking system) database provides information on all NPDES-permitted facilities in the state and allows the public to search for facilities by name, county, or municipality (www.dep.state.pa.us/efacts/).

Some types of facilities and activities with NPDES permits under the DEP Bureau of Watersheds include:

- Discharge from industry
- Discharge of stormwater associated with industrial activities
- Stormwater runoff from construction (greater than one acre disturbance)
- Discharge of stormwater from municipal separate storm sewer systems (MS4s) (see section on stormwater below)
- Publicly owned sewage treatment works
- Discharge from gasoline-contaminated ground water remediation systems
- Single residence sewage treatment plant
- Active mining operations

Examples of facilities that are not permitted, but which affect water quality are: sanitary sewer overflows and illegal sanitary sewer tie-ins to storm drains.

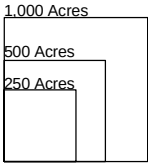
The Potential Sources of Pollution Map indicates the location of oil and gas wells in addition to abandoned mine sites. Any potential source of pollution located within a Susceptible Resource Area or an Exceptional Quality Watershed may have a negative impact on the resources in that area.



- Legend**
- Watershed Boundary
 - County Boundary
 - Municipal Boundary
 - Rivers/Streams/Lakes
 - Highways
 - Major Roads
 - Active Railroads
 - Inactive Railroads
 - Oil and Gas Wells
 - Abandoned Mines

Note: As of 2002 there were no 303d listed waterbodies in the Beaver River Conservation and Management Plan study area.

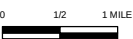
- Subwatersheds**
1. Hamilton Run
 2. Beaver River
 3. McKinley Run
 4. Block House Run
 5. Brady Run
 6. Grimms Run
 7. Walnut Bottom Run
 8. Bennett Run
 9. Wallace Run
 10. Thompson Run
 11. Clarks Run
 12. Stockman Run
 13. Wampum Run
 14. Eckles Run
 15. Snake Run
 16. McKee Run
 17. Jenkins Run
 18. Edwards Run



**POTENTIAL SOURCES
OF POLLUTION**
**BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design,^{INC}

Date: September 2006
1989.05.15r1



Non-Point Sources

Although non-point source pollution is much more difficult to control than point source pollution, there are still efforts throughout the state to prevent and control it. The DEP Bureau of Watershed Management has set up a "Non-Point Source (NPS) Management Program," which consists of action plans that address this type of pollution across the state. Some of the common sources of NPS pollution in Pennsylvania are:

- Abandoned mine drainage (AMD) (see section below)
- Agriculture (runoff of soil that contains fertilizers and excess nutrients)
- Silviculture (soil erosion and sediment loading from forestry operations)
- Construction (runoff of soil into the water which increases chance of flooding)
- Illegal land disposal
- Urban runoff (pesticides, lawn fertilizers, oil, and other chemicals and debris deposited or littered in urban areas)

Two of the most serious types of non-point source pollution are discussed below.

*Abandoned Mine Drainage*¹⁰

While coal was being formed in the Earth millions of years ago, the conditions for its formation also favored the concentration of iron- and sulfur-containing materials that eventually became pyrite (fool's gold). Therefore, pyrite is commonly found along coal seams. As long as pyrite remains undisturbed underground, it generally causes no problem. However, mining can expose pyrite to water and oxygen, which break it down. When this trio reacts, sulfuric acid and a rust-like compound high in iron form "yellowboy." Streams that run orange do so because of yellowboy depositions that tend to smother small aquatic life and destroy the food chain for larger animals, like fish. The acidity formed by the pyrite reaction can also dissolve clay which in turn releases heavy metals like aluminum. Aluminum presents itself as a white deposit and is toxic to fish.

These pollutants can be formed in the underground voids left by deep mining, from the coal refuse piles (slate dumps, gob piles, boney piles) brought to the surface, or by exposing pyrite through strip mining. If sufficient acidity enters a stream, its pH may be significantly lowered resulting in a very toxic environment for aquatic life. Acidity and precipitated metals are the major threats posed by AMD. The nature of AMD contamination varies greatly from site to site, as its formation is dependent upon a variety of factors. Therefore, several parameters are measured when studying AMD:

- Low pH (high acidity), i.e., acid mine drainage
- High metal concentrations (iron, aluminum, and manganese)
- Elevated sulfate levels
- Excessive suspended solids and/or siltation

Because the AMD reaction is dependent on the specifics of the geology and hydrology of the particular site, no two AMD discharges are exactly alike chemically. Suffice it to say, there is not one set way of treating AMD to mitigate its ill effects, but rather many, each depending on the particulars of the discharge. However, a general strategy is usually followed: (1) isolate the AMD, (2) reduce its acidity (if necessary), and (3) cause the iron (and other dissolved metals) to

¹⁰ This section taken from *Monitoring Matters* Volume VII, Number 2, August 2004. Origins of Abandoned Mine Drainage. Pages 1,4. By Deb Simko.

precipitate as a solid in a settling pond and/or wetland. The treated water is then allowed to enter the stream. Whatever the type of AMD, it is this pollution that degrades habitats, causes safety problems, destroys public and private water supplies, ruins the natural aesthetics, and has a negative economic impact.

Stormwater

Stormwater is water from rain or snow that flows over the land and either infiltrates into the ground or drains into nearby streams and can be characterized as both point and non-point source pollution. Natural stormwater runoff from the land or from small construction sites less than one acre is considered to be non-point source pollution because there is no discreet conveyance of the water – it runs over the land and into streams and rivers without controls.

Conversely, stormwater from construction sites larger than one acre or from municipal separate storm sewer systems (MS4s) are considered to be point source pollution, which must be managed and permitted.

Pennsylvania's Stormwater Management Program came out of the Stormwater Management Act (Act 167) of 1978. Under the Program, counties are required to develop stormwater management plans for watersheds within their boundaries. Municipalities then develop ordinances that meet the specifications of the county plans. Plans must be reviewed every five years and include an inventory of both existing and potential characteristics and problems of the area, such as run-off characteristics, soil impacts, and significant obstructions. When construction or other land disturbances take place, the developers must follow the guidelines set forth for stormwater management. Both Beaver and Lawrence Counties have begun developing Act 167 Plans. Contact the County Planning offices for current updates on their progress.

The Clean Water Act established two Phases of the federal Stormwater Program:

Phase I (1992) requires NPDES permits for construction activities that disturb five or more acres of land. Permittees must use best management practices (BMPs) and erosion and sediment control plans to control stormwater runoff from sites.

Phase II (adopted in 2002) requires NPDES permits for construction activities that disturb one to five acres of land. This permit also requires the use of BMPs and erosion and sediment control plans. In addition to the construction permits, Phase II also requires NPDES permits for municipal separate storm sewer systems (MS4s) in urban areas. As part of the permit requirements, the MS4 operators must develop and implement BMPs to manage stormwater and must conduct public outreach. Operators within municipalities that have adopted an Act 167 Plan may already meet some of the requirements of the MS4 NPDES permit if their Act 167 Plan sufficiently addresses water quality issues. Other operators must develop their own stormwater management program or develop an Act 167 Plan to meet permit requirements. These permit requirements must be completed during the five-year permit period (the five year period ends March, 2008). All 20 municipalities in the Beaver County portion of this study are MS4

MS4s (municipal separate storm sewer system)

Normally, sewer systems are separated into a sanitary system (sewage from homes and businesses), and a storm system (drainage from rain or snow). Water from a storm sewer system is not treated and empties into rivers. Municipalities are now required to have permits for these storm sewer discharges. See text on 'Phase II.'

communities. Wayne Township in Lawrence County is the only MS4 in that section of the study area.

Visit www.dep.state.pa.us, keyword “stormwater” for more details.

Impaired Streams and Rivers

While NPDES permits target point source pollution, another approach to targeting all pollution sources, especially non-point, is through the use of Total Maximum Daily Loads (TMDLs). The CWA calls for the development of TMDLs for all waterways that do not meet water quality standards.

According to the PA DEP, TMDLs:

set an upper limit on the pollutant loads that can enter a water body so that the water will meet water quality standards. The Clean Water Act requires states to list all waters that don't meet their water quality standards even after required pollution controls are put into place. For these, the state calculates how much of a substance can be put in the stream without violating the standard and then distribute that quantity among all sources of the pollution on that water body. A TMDL plan includes waste load allocations for point sources, load allocations for non-point sources, and a margin of safety. States must submit TMDLs to the Environmental Protection Agency (EPA).

Assessed waterways that do not meet their designated use must be listed by the state every two years in accordance with Section 303(d) of the CWA, which is the list of impaired streams and rivers. Waterways listed within Section 303(d) are prioritized for TMDL development based on the severity of impairment. The DEP is incorporating them on a watershed basis where local watershed groups actually implement the TMDL Plan and do testing with DEP's assistance. A TMDL for the Beaver River below the Brighton Dam was completed in 2001 for PCB and Chlordane. PCBs (synthetic oils) and Chlordane (pesticide) were banned in 1979 and 1988 respectively. These are chemicals that persist in the environment by becoming embedded in sediment and are easily absorbed into fish tissue. Because there are no longer point sources of these chemicals, restrictions on their release cannot be controlled. It is generally recommended that low-level, spread-out concentrations of PCBs and Chlordanes be left to naturally attenuate. In other words, it is better to let them where they are than to try and remediate them. For example, if the Beaver River bed was dredged, it would most likely cause the chemicals to be disturbed and cause greater harm than if left to settle and be covered by relatively clean sediment.

The Clean Water Act also requires a water quality assessment report (305(b)) on all impaired waters every two years along with the 303(d) list. "This report provides summaries of various water quality management programs including water quality standards, point source control, and non-point source control. It also includes descriptions of programs to protect lakes, wetlands, and groundwater quality."¹¹ Furthermore, the 305(b) report describes the extent to which waterways are supporting their designated uses. For example, if in a particular waterway all designated uses are achieved, the waterway is listed as “fully supporting.”

¹¹ PA DEP www.dep.state.pa.us

In Pennsylvania, DEP has combined the 305 (b) report and the 303 (d) list into one document, "The 2006 Pennsylvania Integrated Water Quality Monitoring and Assessment Report."

The 2006 report lists the following streams that were evaluated for and attained use requirements for aquatic life: Beaver River (also, parts of the river attained use requirements for fish consumption and potable water supply), Blockhouse Run, Brady Run, Eckles Run, Edwards Run, Jenkins Run, McKee Run, Stockman Run, Wallace Run, Snake Run, Bennett Run, and Wampum Run.

The list also includes the following polluted stream not needing a TMDL: Walnut Bottom Run was evaluated for aquatic life and threats to that use included habitat modification, thus causing water and flow variability.

Water Quality Testing

The Army Corps of Engineers oversees the equipment and reporting of stream conditions and water quality at two stations along the Beaver River (Wampum and Beaver Falls). The United States Geological Survey also lists the Army Corps' data on their website.

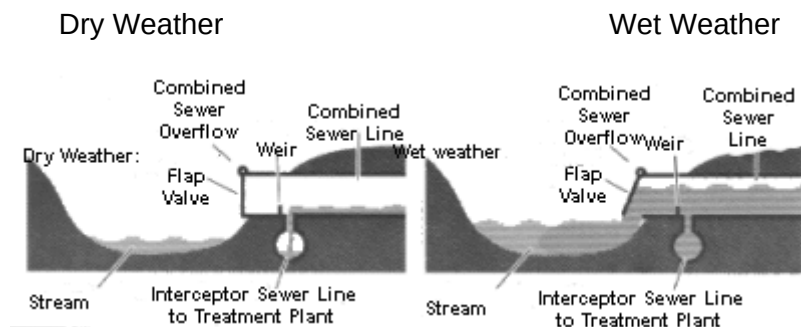
<http://wmw.lrp.usace.army.mil/current/index.html>

<http://pa.water.usgs.gov/current.html>

Sewer Overflows¹²

Combined sewer systems are designed to carry wastewater and stormwater. These are more common in communities with collection systems built before the 1940s. Water and waste from a variety of sources come together in one sewer system and are sent to a water treatment facility. However, during wet weather, the treatment plants cannot handle the capacity of sewage and water, so the pipes overflow to waterways.

When this type of overflow occurs in a combined collection system, it is called a combined sewer overflow (CSO). These were designed with overflow structures to deliberately release excess stormwater and wastewater at capacity. These structures are legal, though they require a permit.



Copyright © 2002 by the Louisville/Jefferson County Metropolitan Sewer District (MSD) Louisville, Kentucky <http://www.msdlouky.org/programs/sso.htm>

¹² The Regionalization Report: An initial study on options for regionalizing the management of sewage collection within the ALCOSAN service area, 3 Rivers Wet Weather, Inc., January 2002.

Separate sanitary sewer systems are designed to carry only wastewater. Stormwater is managed through a different collection system. These systems were required for any new system built after the 1940s.

Sewer pipes are rarely full when wastewater is flowing from homes to the sewage treatment plant. Therefore, groundwater or stormwater can leak into cracked or broken pipes, taking up space that should be used to carry only wastewater. In some instances, stormwater is illegally piped into separate sanitary systems to control the runoff through storm drains in streets, parking lots, and gutters. During dry weather, the sewage systems generally operate effectively. During wet weather, the additional flow exceeds the capacity of the sewers causing the sewage to overflow into creeks, streams, or rivers, creating a large-scale problem.

When this type of overflow occurs in a separate sanitary system, it is called a sanitary sewer overflow (SSO) and may occur in an overflow structure, a structure that is intentionally designed to discharge flow into nearby streams. Occasionally, the overflow can occur in a street from a manhole or in the basements of homes. The overflow structures and unintentional overflows are illegal according to the Clean Water Act. The types of overflows that occur in streets or basements also are illegal.

Sewage Facilities

The Generalized Sanitary Sewer Service Inventory Map shows areas of existing sanitary sewer infrastructure by supplier within the Study Area. Sanitary sewer service utilizes an underground carriage system for transporting sewage from houses or industry to a treatment facility. Based upon county data, existing sewer service is concentrated in the southern portion of the Beaver River Valley, near its confluence with the Ohio River. In general, the development potential of an area is largely dependent on the availability of sanitary sewer infrastructure. It is therefore likely that most types of development will occur within close proximity to sewer lines or in areas where sewer extensions have been planned. At the same time, all sewer systems are limited by capacity, either within the piping system or within the sewage treatment facility. As a result, a strategic sewer service expansion program is an effective way to manage growth in a particular area.

Sanitary sewer service encompasses approximately 20 percent, of the Study Area. Colors are assigned to the nine individual municipal sanitary sewer service providers that manage public infrastructure to these communities. The majority of sanitary sewer service is located in proximity to the Beaver River and Ohio River confluence near Beaver Borough, Bridgewater Borough and Beaver Falls City. North of this area, communities have more limited access to available sanitary sewer service and smaller pockets of intense development are present as well as overall patterns of lower intensity development.

Another depiction of the relationship between existing land use and sanitary sewer service is on the Generalized Land Coverage and Sanitary Sewer Service Analysis. The map illustrates portions of the study area that have the necessary infrastructure service to support development. The specific capacity, age and adequacy of these existing systems to support any new development should be continually monitored in order to ensure development and public service planning efforts are coordinated.

The Sewage Facilities Program (Act 537) was established in 1966 to ensure proper oversight and permitting of onlot disposal systems (OLDS), which are individual or community septic

systems. The program is administered by local municipalities or county health departments, known as local agencies. Those agencies must hire a Sewage Enforcement Officer (SEO) to oversee the permitting process for OLDS (the PA DEP offers only technical and financial assistance). The local agencies must also have an Act 537 comprehensive plan that addresses current and future sewage disposal needs.

The following list shows when each municipality had their 537 Plan approved.¹³

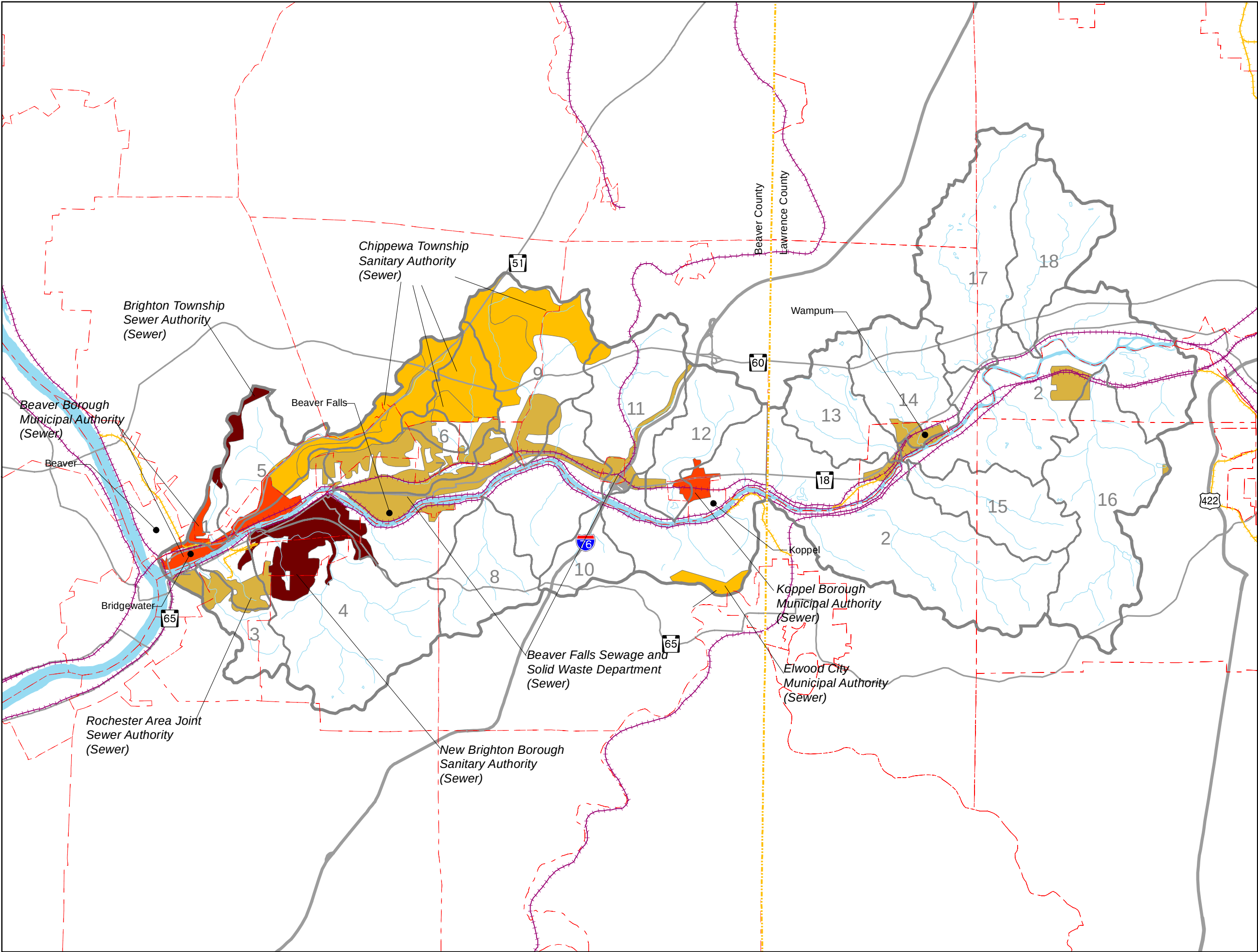
Beaver County

Beaver Borough	1/1971
Beaver Falls City	1/1977
Big Beaver Borough	7/2003
Bridgewater Borough	8/1971
Brighton Township	1/1971
Chippewa Township	7/1991
Daugherty Township	1/1971
Eastvale Borough	1/1971
Fallston Borough	1/1971
Homewood Borough	1/1997
Koppel Borough	7/2003
New Brighton Borough	NA
North Sewickley Township	7/2003
Patterson Heights Borough	1/1971
Patterson Township	4/1985
Pulaski Township	1/1971
Rochester Borough	1/1971
Rochester Township	1/1971
West Mayfield Borough	1/1971
White Township	7/2004

Lawrence County

New Beaver Borough	11/1971
North Beaver Township	5/1999
Shenango Township	1/1966
Taylor Township	10/1971
Wampum Borough	9/1971
Wayne Township	7/2004

¹³ From www.dep.state.pa.us

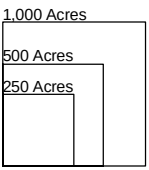


- Legend**
- Study Area Boundary
 - Watershed Boundary
 - County Boundary
 - Municipal Boundary
 - Rivers/Streams/Lakes
 - Highways
 - Major Roads
 - Active Railroads
 - Inactive Railroads

Sanitary Sewer Service Authorities

1. Brighton Township Sewer Authority	189 ac.
2. Beaver Borough Municipal Authority	642 ac.
3. Rochester Area Joint Sewer Authority	536 ac.
4. Chippewa Township Sanitary Authority	4,095 ac.
5. New Brighton Borough Sanitary Authority	1,264 ac.
6. Beaver Falls Sewage and Solid Waste Department	3,263 ac.
7. Elwood City Municipal Authority	187 ac.
8. Koppel Borough Municipal Authority	187 ac.
9. Other	626 ac.

- Sub-Watersheds**
1. Hamilton Run
 2. Beaver River
 3. McKinley Run
 4. Block House Run
 5. Brady Run
 6. Grimms Run
 7. Walnut Bottom Run
 8. Bennett Run
 9. Wallace Run
 10. Thompson Run
 11. Clarks Run
 12. Stockman Run
 13. Wampum Run
 14. Eckles Run
 15. Snake Run
 16. McKee Run
 17. Jenkins Run
 18. Edwards Run

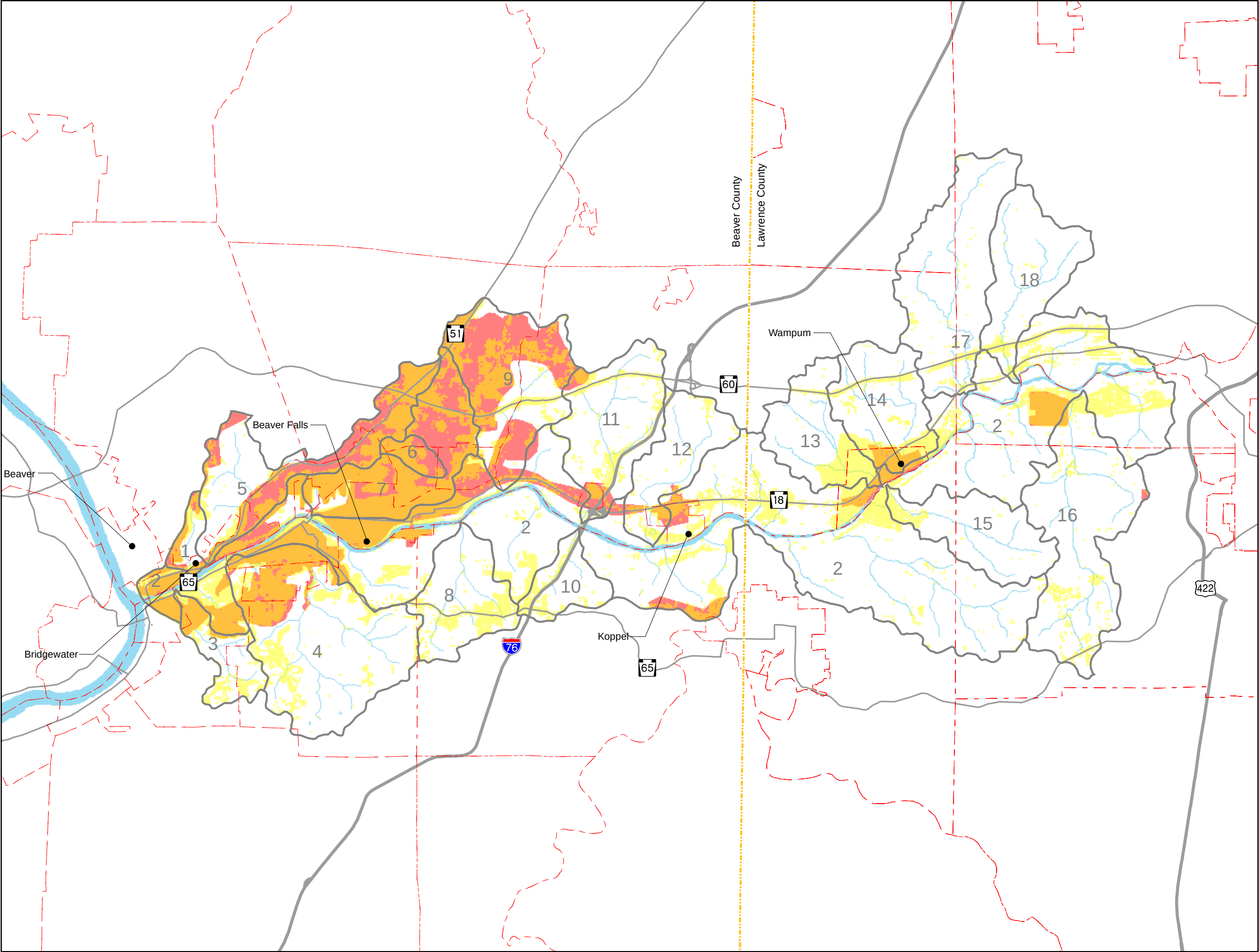


**GENERALIZED SANITARY SEWER SERVICE INVENTORY
BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: July 2008
1989.05.11r2





Legend

Study Area Boundary

Watershed Boundary

County Boundary

Municipal Boundary

Rivers/Streams/Lakes

Major Highways

Major Roads

Land Coverage

Urban Area with Sanitary Sewer Service

Non-urban Area with Sanitary Sewer Service

Urban Area without Sanitary Sewer Service

Non-urban Area without Sanitary Sewer Service

Note:
1. Urban land is comprised of low density urban areas and high density urban areas as classified by Penn State University.
2. Where coverage data along the study area boundary was not available, area appears as non-urban land coverage.

- Sub-Watersheds**
1. Hamilton Run

2. Beaver River

3. McKinley Run

4. Block House Run

5. Brady Run

6. Grimms Run

7. Walnut Bottom Run

8. Bennett Run

9. Wallace Run

10. Thompson Run

11. Clarks Run

12. Stockman Run

13. Wampum Run

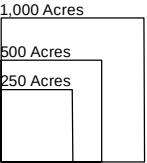
14. Eckles Run

15. Snake Run

16. McKee Run

17. Jenkins Run

18. Edwards Run



Fish Consumption Advisories

Due to the presence of pollutants in the streams and rivers, the PA Fish and Boat Commission (PFBC) and the DEP issue fish consumption advisories each year. They provide recreational fishermen with guidelines of how many fish they may eat in a certain time period based on the severity of the pollution in fish tissue. There is also a general statewide fish consumption advisory: eat no more than one meal (1/2 pound) per week of state-caught sport fish.

Table 4-7 2007 Fish Consumption Advisories			
Water Course	Fish Species	Meal Frequency*	Contaminant
Beaver River confluence to New Brighton Dam	Carp/ Channel Catfish	6 meals/year	PCB
	Smallmouth Bass	1 meal/month	
Beaver River at New Brighton Dam to mouth	Carp/ Channel Catfish	Do not eat	PCB
Source: www.fish.state.pa.us * One meal is a half pound of fish for a 150-lb person.			

Contact Information

To report suspected water quality violations, contact one of the following:

In Beaver County, contact the Southwest Regional Office of the PA Department of Environmental Protection at (412) 442-4184

In Lawrence County, contact the Northwest Regional Office of the DEP at (814) 332-6816

Water Supply

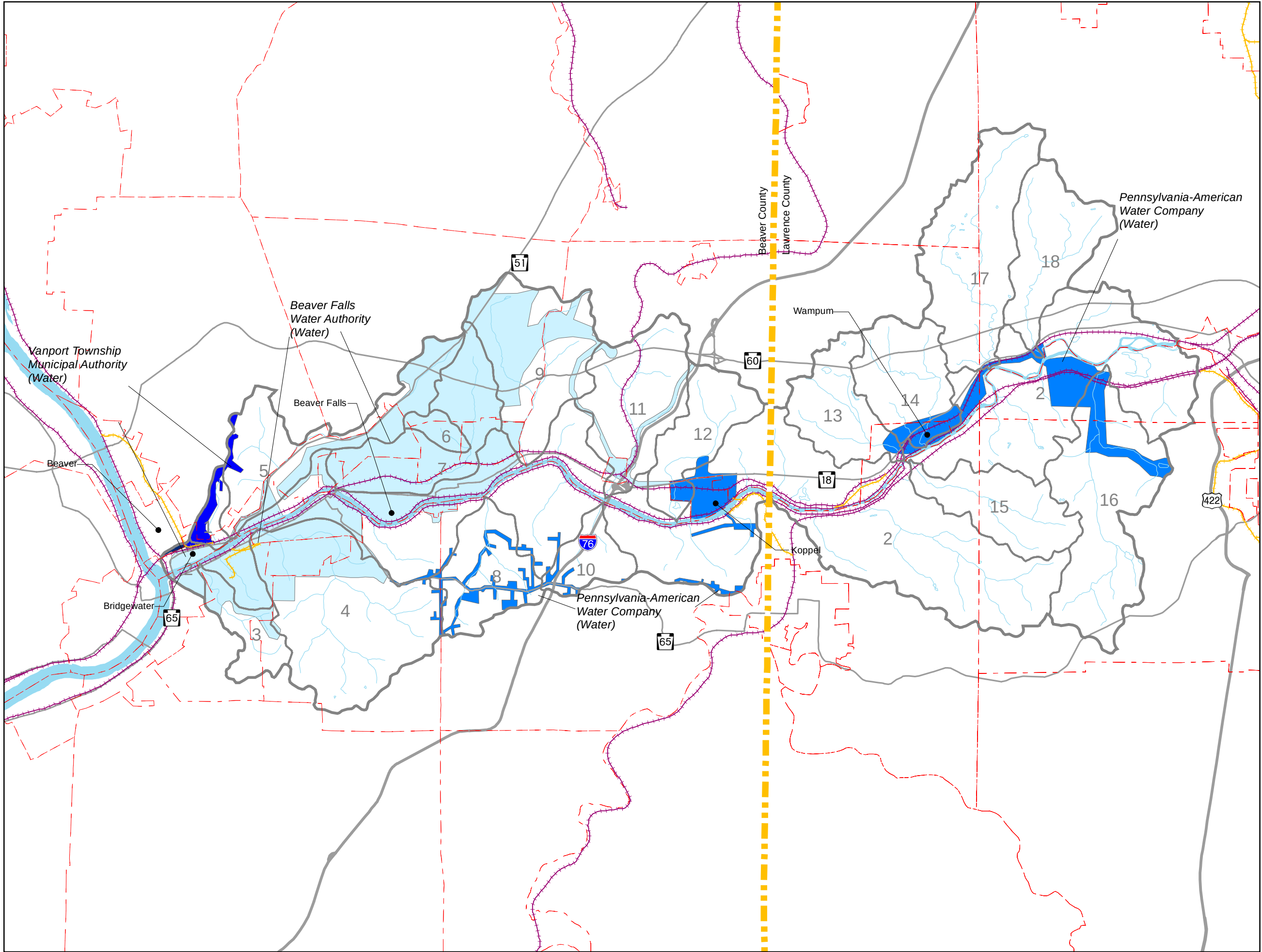
Water Services Inventory

The Generalized Water Service Inventory map indicates the widespread availability of public water by provider in the southern half of the Study Area, with smaller systems scattered throughout the rest of the region. Like sewer, water plays an important role in all but the least intensive land developments and is frequently located in the same locations. Public water service is almost always available to those who live in urban and suburban areas and is rarely available in remote areas. Whereas sewer systems are usually limited by transport and treatment capacities, water systems are always dependent on an adequate supply. Thus, water quality, storage, transport and watershed protection are all important factors to consider when analyzing a public water system.

On the map, colors are assigned to the four individual municipal water service providers - supplying 23 percent of the Study Area with water. The Beaver Falls Water Authority supplies the majority of the water service, primarily near the confluence of the Beaver and Ohio Rivers.

Smaller service areas exist in the northern portion of the study area, but the majority of the study area north of the Ohio River and Beaver River confluence lacks a public water service provider.

Heavily populated areas in close proximity to the Beaver and Ohio River confluence are provided with water service as illustrated on the Generalized Land Coverage by Water Service Analysis Map. As the distance increases from the confluence of the Beaver and Ohio Rivers, urban areas and available water service decreases. The map illustrates which portions of the study area have the necessary infrastructure service that typically supports more intense areas of development. In some locations within the Project Corridor, existing infrastructure service is also generally accessible to lands that have minimal and/or no development, providing opportunity to readily support future development.

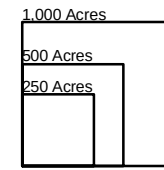


- Legend**
- Study Area Boundary
 - Watershed Boundary
 - County Boundary
 - Municipal Boundary
 - Rivers/Streams/Lakes
 - Highways
 - Major Roads
 - Active Railroads
 - Inactive Railroads

Water Service Authorities

1. Vanport Township Municipal Authority	271 ac.
2. Beaver Falls Water Authority	9,665 ac.
3. Pennsylvania-American Water Company	3,287 ac.
4. Beaver Borough Municipal Authority	228 ac.

- Sub-Watersheds**
1. Hamilton Run
 2. Beaver River
 3. McKinley Run
 4. Block House Run
 5. Brady Run
 6. Grimms Run
 7. Walnut Bottom Run
 8. Bennett Run
 9. Wallace Run
 10. Thompson Run
 11. Clarks Run
 12. Stockman Run
 13. Wampum Run
 14. Eckles Run
 15. Snake Run
 16. McKee Run
 17. Jenkins Run
 18. Edwards Run

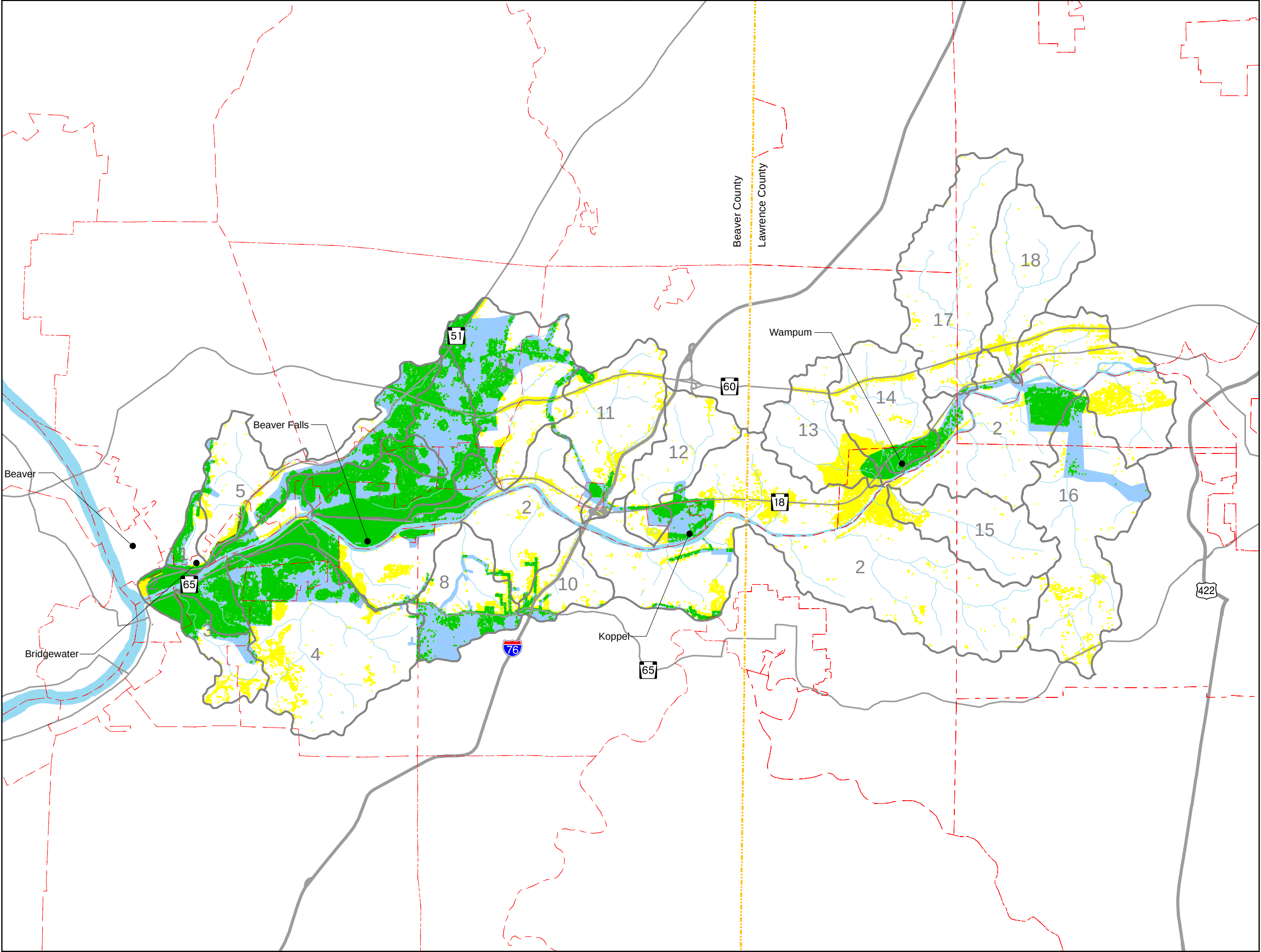


**GENERALIZED WATER
SERVICE INVENTORY
BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: July 2008
1989.06.12r3





Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundary
- Rivers/Streams/Lakes
- Major Highways
- Major Roads

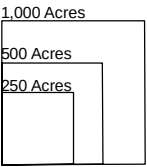
Land Coverage

- Urban Area with Water Service
- Non-urban Area with Water Service
- Urban Area without Water Service
- Non-urban Area without Water Service

Note:
1. Urban land is comprised of low density urban areas and high density urban areas as classified by Penn State University.
2. Where coverage data along the study area boundary was not available, area appears as non-urban land coverage.

Sub-Watersheds

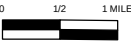
1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run



**GENERALIZED LAND COVERAGE
AND WATER SERVICE ANALYSIS
BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.06.05r2



Pennsylvania Source Water Assessment Program

In August of 1996, Congress amended the Safe Drinking Water Act to include provisions for drinking water supply assessments. One year later, the EPA issued a guidance document for states to begin their Source Water Assessment and Protection (SWAP) program, the main goals of which are the prevention of drinking water contamination and citizen involvement in the clean-up and pollution prevention process.

The DEP, along with the consulting firm Spotts, Stevens, and McCoy, assessed all the drinking water sources in the state. The assessments include one square mile surrounding all groundwater systems serving a population of 3,300 or more, small surface water systems with small, forested watersheds, and large surface water systems. The studies looked at the prime contributors to water pollution and how they affect the water quality of the drinking water source. Each of the assessments will be used to determine what preventative steps are required (by municipalities, water suppliers, and the government) to protect drinking water systems.¹⁴

The assessments summarized the greatest potential threats to water quality. While these threats may not be present within the boundaries of the study corridor, they still affect water quality as an upstream source. Summaries of source water facility reports are available on the DEP's website (www.dep.state.pa.us) under the Source Water and Groundwater Protection Page heading. As of June 2008, only the Beaver Falls Municipal Authority and the Beaver Borough Municipal Authority have their reports posted.

The source water reports only assess the quality of the source of water (ground or surface). To obtain information on the quality of tap water, contact the local water authority about their annual consumer confidence reports. These reports must be distributed each year to customers of water suppliers and will list the water source, average results of water quality tests, whether the tests showed that the water met appropriate standards, and how consumers can help to protect their water supply.

Act 220 – The Water Resources Planning Act

Act 220 requires the state to update the State Water Plan. The update will include an in-depth look at how our water resources are being used. In other words, it will be a “water withdrawal and use registration and reporting program. This inventory will allow the DEP to determine how much water Pennsylvania has, how much is being used, and how much will be available in the future.”¹⁵ Water usage must be reported annually by all public water suppliers, hydropower stations, and individuals or businesses that withdraw more than 10,000 gallons per day (over a 30-day period). Any of those entities that withdraw over 50,000 gallons per day must meter their water use. Usages between 10,000 and 50,000 gallons must be estimated (DEP offers methods for estimating use). Homeowners are not required to monitor or report water usage (typically, households use 300 gallons per day).

Statewide Water Resources Committees have been established for the major river basins in the state (this Plan falls within the Ohio River Basin) and will “oversee the creation of water resources plans on the local level.”

¹⁴ <http://www.dep.state.pa.us/dep/deputate/watermgt/wc/subjects/srcprot/sourceassessment/default.htm>

¹⁵ www.dep.state.pa.us

Groundwater

Water filtering through the soil moves first through an unsaturated zone where the spaces (pores) between solid particles or rocks contain both air and water. At this stage water is called soil water and some of it will be taken up by plants. The rest of it continues, pulled by gravity, in a generally downward path and eventually reaches the zone of saturation. Here the pore spaces are completely filled with water; this is groundwater. The top of the zone of saturation is called the water table. Rock or soil layers within the zone that can readily store and transmit usable amounts of water are called aquifers.¹⁶

Since much of the water supply for western Pennsylvanians comes from surface water, not much thought is given to groundwater. It is important to know, however, that groundwater provides about two thirds of water to streams, lakes, and wetlands (this discharge of groundwater to surface water is called “base flow”). In fact, most of Pennsylvania’s fresh water comes from groundwater, which is commonly used for agriculture, industry, mining, domestic, and commercial uses.

Biological Resources

Wildlife

Mammals

There are 70 mammals known to live or have lived in Pennsylvania. The mammals listed in Table 4-8 can be found in Beaver and Lawrence Counties based on published data.

Table 4-8 Mammals in Beaver and Lawrence Counties	
Common Name	Scientific Name
White-Tailed deer	<i>Odocoileus virginianus</i>
Striped skunk	<i>Mephitis mephitis</i>
Mink	<i>Mustela vision</i>
Weasel	<i>Mustela spp.</i>
Raccoon	<i>Procyon lotor</i>
Red fox	<i>Vulpes vulpes</i>
Gray fox	<i>Urocyon cinereoargenteus</i>
Coyote	<i>Canis latrans</i>
Meadow jumping mouse	<i>Zapus hudsonius</i>
Woodland jumping mouse	<i>Napaeozapus insignius</i>
Bog lemming	<i>Synaptomys cooperi</i>
Muskrat	<i>Ondatra zibethicus</i>
Voies	<i>Microtus spp</i>

¹⁶ Groundwater: A Primer for Pennsylvanians. Water Resources Education Network.

Deer mouse	<i>Peromyscus maniculatus</i>
White-footed mouse	<i>Peromyscus leucopus</i>
Woodrat	<i>Neotoma floridana</i>
Beaver	<i>Castor canadensis</i>
Southern flying squirrel	<i>Glaucomys volans</i>
Red squirrel	<i>Tamiasciurus hudsonicus</i>
Eastern chipmunk	<i>Tamias striatus</i>
Fox squirrel	<i>Sciurus niger</i>
Gray squirrel	<i>Sciurus carolinensis</i>
Groundhog	<i>Marmota monax</i>
Eastern cottontail	<i>Sylvilagus floridanus</i>
Eastern pipistrelle	<i>Pipistrellus subflavus</i>
Indiana bat*	<i>Myotis sodalis</i>
Little brown bat	<i>Myotis lucifugus</i>
Northern long-eared bat**	<i>Myotis septentrionalis</i>
Hoary bat	<i>Lasiurus cinereus</i>
Red bat	<i>Lasiurus borealis</i>
Silver haired bat **	<i>Lasionycteris noctivagans</i>
Big brown bat	<i>Eptesicus fuscus</i>
Hairy tailed mole	<i>Parascalops breweri</i>
Star-nosed mole	<i>Condylura cristata</i>
Pygmy shrew	<i>Sorex hoyi</i>
Smoky shrew	<i>Sorex fumeus</i>
Masked shrew	<i>Sorex cinereus</i>
Least shrew*	<i>Cryptotis parva</i>
Short-tailed shrew	<i>Blarina brevicauda</i>
Opossum	<i>Didelphis virginiana</i>
House mouse	<i>Mus musculus</i>
Norway rat	<i>Rattus norvegicus</i>
Source: www.carnegiemuseums.org (April 2005)	
* Endangered	
** Rare	

Pennsylvania's mammals and birds are managed by the PA Game Commission (PGC). "Wildlife management is the process used to manage game and other wildlife populations and includes: monitoring wildlife populations, establishing laws and regulations, setting seasons and bag limits, making habitat improvements, providing outright protection, informing and educating the public (www.pgc.state.pa.us)." They also reintroduce species and manage endangered and threatened species. More information about threatened and endangered species and the Game Commission are found later in this chapter.

Several organizations are working toward conservation and stewardship of wildlife in the study area. The Important Mammal Areas Project (IMAP) is a “voluntary grass roots project with two goals: 1) to designate areas in Pennsylvania that are important for mammal conservation, and 2) to help people learn more about mammals and their habitats.” The project is a partnership of the PA Wildlife Federation, National Wildlife Federation, Indiana University of Pennsylvania, Carnegie Museum of Natural History, PA Game Commission, PA Federation of Sportsmens Clubs, and the Mammal Technical Committee of the Pennsylvania Biological Survey.¹⁷

Important Mammal Areas (IMA) are geographic areas that host either game or non-game species of concern and meet one or more of several criteria: diverse or unique mammals, high density population, species listed as endangered or threatened, species that are declining or vulnerable, and/or important for public education. Currently, there are 45 IMAs in Pennsylvania.

There is one IMA in the Study Area. Because the species for which the site has been nominated is of special concern, and the property where it was found is privately owned, details about them, or the exact locations cannot be released. The IMAP partners are currently working on completing conservation assessments on all sites in western Pennsylvania. Once those assessments are complete, a conservation/stewardship plan may be written for each.

Birds

Information about birds that can be found in Beaver and Lawrence Counties can be found in the *Atlas of Breeding Birds in Pennsylvania*, published in 1992. The atlas is available online at www.carnegiemnh.org/atlas. Information for the atlas was collected from volunteers and experts throughout the state. The Carnegie Museum of Natural History is sponsoring an update to the atlas. The *Second Pennsylvania Breeding Bird Atlas* should be completed in 2009.

Table 4-9 contains a compilation of the birds recorded by volunteers during the annual Christmas Bird Count in the Beaver Valley.

Table 4-9 Species List for Beaver Valley Christmas Bird Count 1987 – 2004	
Common Name	Scientific Name
Pied-billed Grebe	<i>Podilymbus podiceps</i>
American White Pelican	<i>Pelecanus erythrorhynchos</i>
Double-crested Cormorant	<i>Phalacrocorax auritus</i>
Great Blue Heron (Blue form)	<i>Ardea herodias</i>
White-fronted Goose	<i>Anser albifrons</i>
Snow Goose	<i>Chen caerulescens</i>
Canada Goose	<i>Branta canadensis</i>
Tundra Swan	<i>Cygnus columbianus</i>
Wood Duck	<i>Aix sponsa</i>
American Wigeon	<i>Anas Americana</i>

¹⁷ www.pawildlife.org

American Black Duck	<i>Anas rubripes</i>
Mallard	<i>Anas platyrhynchos</i>
Northern Pintail	<i>Anas acuta</i>
American Green-winged Teal	<i>Anas crecca</i>
Canvasback	<i>Aythya valisineria</i>
Redhead	<i>Aythya americana</i>
Surf Scoter	<i>Melanitta perspicillata</i>
Common Goldeneye	<i>Bucephala clangula</i>
Hooded Merganser	<i>Lophodytes cucullatus</i>
Common Merganser	<i>Mergus merganser</i>
Red-breasted Merganser	<i>Mergus serrator</i>
Ruddy Duck	<i>Oxyura jamaicensis</i>
American Bald Eagle	<i>Haliaeetus leucocephalus</i>
Northern Harrier	<i>Circus cyaneus</i>
Sharp-shinned Hawk	<i>Accipiter striatus</i>
Cooper's Hawk	<i>Accipiter cooperii</i>
Northern Goshawk	<i>Accipiter gentilis</i>
Red-shouldered Hawk	<i>Buteo lineatus</i>
Red-tailed Hawk	<i>Buteo jamaicensis</i>
American Kestrel	<i>Falco sparverius</i>
Ring-necked Pheasant	<i>Phasianus colchius</i>
Ruffed Grouse	<i>Bonasa umbellus</i>
Wild Turkey	<i>Meleagris gallopavo</i>
American Coot	<i>Fulica americana</i>
Killdeer	<i>Charadrius vociferous</i>
Common Snipe	<i>Gallinago gallinago</i>
Ring-billed Gull	<i>Larus delawarensis</i>
Herring Gull	<i>Larus argentatus</i>
Great Black-backed Gull	<i>Larus marinus</i>
Rock Pigeon	<i>Columba livia</i>
Mourning Dove	<i>Zenaida macroura</i>
Eastern Screech-Owl	<i>Otus asio</i>
Barred Owl	<i>Strix varia</i>
Great Horned Owl	<i>Bubo virginianus</i>
Short-eared Owl	<i>Asio flammeus</i>
Belted Kingfisher	<i>Ceryle alcyon</i>
Red-headed Woodpecker	<i>Melanerpes erythrocephalus</i>
Yellow-bellied Sapsucker	<i>Sphyrapicus varius</i>

Downy Woodpecker	<i>Picoides pubescens</i>
Hairy Woodpecker	<i>Picoides villosus</i>
Northern (Yellow-shafted) Flicker	<i>Colaptes auratus</i>
Pileated Woodpecker	<i>Dryocopus pileatus</i>
Blue Jay	<i>Cyanocitta cristata</i>
American Crow	<i>Corvus brachyrhynchos</i>
Horned Lark	<i>Eremophila alpestris</i>
Carolina Chickadee	<i>Parus carolinensis</i>
Black-capped Chickadee	<i>Parus atricapillus</i>
Tufted Titmouse	<i>Parus bicolor</i>
Red-breasted Nuthatch	<i>Sitta canadensis</i>
White-breasted Nuthatch	<i>Sitta carolinensis</i>
Brown Creeper	<i>Certhia americana</i>
Carolina Wren	<i>Thryothorus ludovicianus</i>
Winter Wren	<i>Troglodytes troglodytes</i>
Golden-crowned Kinglet	<i>Regulus satrapa</i>
Ruby-crowned Kinglet	<i>Regulus calendula</i>
Eastern Bluebird	<i>Sialia sialis</i>
Hermit Thrush	<i>Catharus guttatus</i>
American Robin	<i>Turdus migratorius</i>
Northern Mockingbird	<i>Mimus polyglottos</i>
European Starling	<i>Sturnus vulgaris</i>
Cedar Waxwing	<i>Bombycilla cedrorum</i>
Yellow Warbler	<i>Dendroica petechia</i>
Yellow-rumped (Myrtle) Warbler	<i>Dendroica coronata</i>
Eastern Towhee	<i>Pipilo erthrophthalmus</i>
American Tree Sparrow	<i>Spizella arborea</i>
Chipping Sparrow	<i>Spizella passerine</i>
Field Sparrow	<i>Spizella pusilla</i>
Eastern Fox Sparrow	<i>Passerella iliaca</i>
Song Sparrow	<i>Melospiza melodia</i>
Swamp Sparrow	<i>Melospiza georgiana</i>
White-throated Sparrow	<i>Zonotrichia albicollis</i>
Dark-eyed (Slate-colored) Junco	<i>Junco hyemalis</i>
Snow Bunting	<i>Plectrophenax nivalis</i>
Northern Cardinal	<i>Cardinalis cardinalis</i>
Red-winged Blackbird	<i>Agelaius phoeniceus</i>
Common Grackle	<i>Quiscalus quiscula</i>

Brown-headed Cowbird	<i>Molothrus ater</i>
Purple Finch	<i>Carpodacus purpureus</i>
House Finch	<i>Carpodacus mexicanus</i>
Red Crossbill	<i>Loxia curvirostra</i>
Common Redpoll	<i>Carduelis flammea</i>
Pine Siskin	<i>Carduelis pinus</i>
American Goldfinch	<i>Carduelis tristis</i>
Evening Grosbeak	<i>Coccothraustes vespertinus</i>
House Sparrow	<i>Passer domesticus</i>

Fish

The Pennsylvania Fish and Boat Commission protects and manages the state's aquatic resources, these include game and non-game fish, as well as amphibians, reptiles and aquatic invertebrates.

Examples of fish that have been found in the study area are listed in the following tables.

Table 4-10 Beaver River Fish	
Common Name	Scientific Name
Banded Killifish	<i>Fundulus diaphanus</i>
Black Crappie	<i>Pomoxis nigromaculatus</i>
Bluegill	<i>Lepomis macrochirus</i>
Bluntnose Minnow	<i>Pimephales notatus</i>
Brown Bullhead	<i>Ameiurus nebulosus</i>
Channel Catfish	<i>Ictalurus punctatus</i>
Common Carp	<i>Cyprinus carpio</i>
Common Shiner	<i>Luxilus cornutus</i>
Creek Chub	<i>Semotilus atromaculatus</i>
Gizzard Shad	<i>Dorosoma cepedianum</i>
Golden Redhorse	<i>Moxostoma erythrurum</i>
Goldfish	<i>Carassius auratus</i>
Largemouth Bass	<i>Micropterus salmoides</i>
Logperch	<i>Percina caprodes</i>
Northern Hog Sucker	<i>Hypentelium nigricans</i>

Pumpkinseed	<i>Lepomis gibbosus</i>
Rock Bass	<i>Ambloplites rupestris</i>
Sand Shiner	<i>Notropis stramineus</i>
Sauger	<i>Sander canadense</i>
Sheepshead Minnow	<i>Cyprinodon variegatus</i>
Shorthead Redhorse	<i>Moxostoma macrolepidotum</i>
Smallmouth Bass	<i>Micropterus dolomieu</i>
Spotfin Shiner	<i>Cyprinella spiloptera</i>
Walleye	<i>Sander vitreus</i>
White Bass	<i>Morone chrysops</i>
White Crappie	<i>Pomoxis annularis</i>
White Sucker	<i>Catostomus commersoni</i>
White X Striped Bass	<i>White x striped bass</i>
Yellow Bullhead	<i>Ameiurus natalis</i>
Yellow Perch	<i>Perca flavescens</i>
Source: Western Pennsylvania Conservancy, 1983 data	

Table 4-11
Beaver River Fish Observed While Electrofishing
August 24, 2004 and September 1, 2004

Common Name	Scientific Name
Longnose Gar	<i>Lepisosteus osseus</i>
Mooneye	<i>Hiodon tergisus</i>
Gizzard Shad	<i>Dorosoma cepedianum</i>
Common Carp	<i>Cyprinus carpio</i>
Spotfin Shiner	<i>Cyprinella spiloptera</i>
Emerald Shiner	<i>Notropis atherinoides</i>
Quillback	<i>Carpodes carpio</i>
Smallmouth Buffalo	<i>Ictiobus bubalus</i>
River Redhorse	<i>Moxostoma carinatum</i>
Shorthead Redhorse	<i>Moxostoma macrolepidotum</i>
Channel Catfish	<i>Ictalurus punctatus</i>
Flathead Catfish	<i>Pylodictis olivaris</i>
Muskellunge	<i>Esox masquinongy</i>

White Bass	<i>Morone chrysops</i>
Striped Bass	<i>Morone saxatilis</i>
Bluegill	<i>Lepomis macrochirus</i>
Smallmouth Bass	<i>Micropterus salmoides</i>
Sauger	<i>Stizostedion canadense</i>
Walleye	<i>Stizostedion vitreum</i>
Freshwater Drum	<i>Aplodinotus grunniens</i>
Source: Southwest Regional Office DEP	

Table 4-12 Brady Run Fish	
Common Name	Scientific Name
Banded Killifish	<i>Fundulus diaphanus</i>
Blacknose Dace	<i>Rhinichthys atratulus</i>
Bluegill	<i>Lepomis macrochirus</i>
Bluntnose Minnow	<i>Pimephales notatus</i>
Brown Trout	<i>Salmo trutta</i>
Central Stoneroller	<i>Campostoma anomalum</i>
Channel Catfish	<i>Ictalurus punctatus</i>
Common Carp	<i>Cyprinus carpio</i>
Common Shiner	<i>Luxilus cornutus</i>
Creek Chub	<i>Semotilus atromaculatus</i>
Fantail Darter	<i>Etheostoma flabellare</i>
Golden Redhorse	<i>Moxostoma erythrurum</i>
Johnny Darter	<i>Etheostoma nigrum</i>
Largemouth Bass	<i>Micropterus salmoides</i>
Mottled Sculpin	<i>Cottus bairdi</i>
Mountain Brook Lamprey	<i>Ichthyomyzon greeleyi</i>
Northern Hog Sucker	<i>Hypentelium nigricans</i>

Rainbow Darter	<i>Etheostoma caeruleum</i>
Rainbow Trout	<i>Oncorhynchus mykiss</i>
Smallmouth Bass	<i>Micropterus dolomieu</i>
White Sucker	<i>Catostomus commersoni</i>
Source: Western Pennsylvania Conservancy, 1990 and 1998 data	

The PA Fish and Boat Commission maintains a stocking program for the Beaver River to enhance the sport fishery. Table 4-13 lists the 2006 stocking activities.

Table 4-13 2006 Warm/Cool Water Stocking			
Water body	Limits	Species	Lifestage
Beaver River, Beaver County	Connoquenessing Creek to Ohio River	Channel catfish, Sauger, Tiger Muskellunge	fingerling
Beaver River, Lawrence County	Mahoning- Shanango Rivers to Connoquenessing Creek	Tiger Muskellunge	fingerling
Source: www.fish.state.pa.us , August 2006			

In addition, the South Branch of Brady's Run is stocked with brown and rainbow trout prior to the fishing season.

Macroinvertebrates

Macroinvertebrates are also important to consider when discussing river health because they are a food source for other organisms. Although they are near the bottom of the food chain, their abundance indicates a healthy food supply for the rest of the chain. The presence of certain macroinvertebrates is an indicator of water quality. When pollution intolerant species are found in the rivers, it is a good sign of healthy waterways. The more pollution tolerant species present and the lack of intolerant species, mean that water quality may be degraded.

Vegetation

A Natural Heritage Inventory (NHI) "identifies and maps a county's most significant natural places by investigating plant and animal species and natural communities that are unique or uncommon in the county. Areas important for wildlife habitat and scientific study are also

included.”¹⁸ The NHI can be used as a tool by government officials, public and private organizations, and developers who need information about the natural resources of the county. The listing of a location in an NHI does not afford it any special protection.

According to the *1993 Beaver County Natural Heritage Inventory* (NHI), prepared by the Western Pennsylvania Conservancy, there are a variety of vegetation types across the county. The majority of forests on slopes and uplands contain sugar maple (*Acer saccharum*), white oak (*Quercus alba*), red oak (*Quercus rubra*), hickories (*Carya spp.*), American beech (*Fagus grandifolia*), American basswood (*Tilia americana*), and white ash (*Fraxinus americana*). Eastern hemlock (*Tsuga canadensis*) occurs in smaller stream valleys, northern slopes and protected areas. Floodplain forests along streams support sycamore (*Platanus occidentalis*), silver maple (*Acer saccharinum*), American elm (*Ulmus Americana*), and box elder (*Acer negundo*). All of the original forests were logged, therefore these are second growth forests. In addition to the forested communities, the NHI identified the presence of wetland communities in the county.

The *2002 Lawrence County Natural Heritage Inventory* notes that dry upland areas are dominated by oaks (*Quercus spp.*). Lower slopes contain communities of red and white oak, sugar maple, beech, black birch, slippery elm (*Ulmus rubra*), basswood (*Tilia Americana*), white ash (*Fraxinus Americana*), tulip tree (*Liriodendron tulipifera*), and cucumber tree (*Magnolia accuminata*). On northern slopes or along lower sections of stream valleys are hemlock, yellow birch (*Betula allegheniensis*), beech (*Fagus grandifolia*), and rosebay (*Rhododendron maximum*).

The Forest Density Map shows the number of trees per acre for defined areas throughout the Study Area. The densest areas, indicated by darker shades of green, are characterized by thick vegetation and a continuous canopy. The least dense areas, colored in lighter greens, account for minimally planted urban areas as well as agricultural fields. Respective of a slight trend toward denser forests in the northern reaches of the Study Area, the general level of forest density remains similar throughout.

The DCNR Bureau of Forestry offers assistance to municipalities regarding planning for stewardship, conservation, and proper use of forests and other related natural resources. The Bureau employs experts in forestland conservation practices that will provide sound, impartial advice to communities wishing to conserve and enhance their natural resources and maximize the myriad benefits they provide. The Bureau also provides advice to municipalities on various forms of green space and green infrastructure, including the planting and care of trees in developed or developing areas.

The Forest Stewardship Program is a federal and state partnership that assists landowners in the completion of plans focusing on sustainable management of the forest and its related natural resources. The intention of the program is to provide sound management and continuing care for Pennsylvania's private forest resources in the future. Limiting cost share funding is currently available to offset the cost of preparing a Forest Stewardship Plan. Plans must be written by approved plan writers. Information on this opportunity can be obtained at the nearest Bureau of Forestry Field office in Clarion, PA. See www.dcnr.state.pa.us for more information.

¹⁸ Lawrence County Natural Heritage Inventory, Western Pennsylvania Conservancy.

Both woodlands and agriculture areas serve as functional environments for humans as well as natural habitats for wildlife. The Study Area has ample woodland and agriculture areas as shown in the Woodlands and Agriculture Areas Map. The western section of the study area is primarily comprised of agricultural areas, while the eastern section of the study area is primarily comprised of woodland areas.

Habitable woodlands are those lands that contain forest cover with minimum of 300' depth from their center and where such land is a minimum of 250 acres.

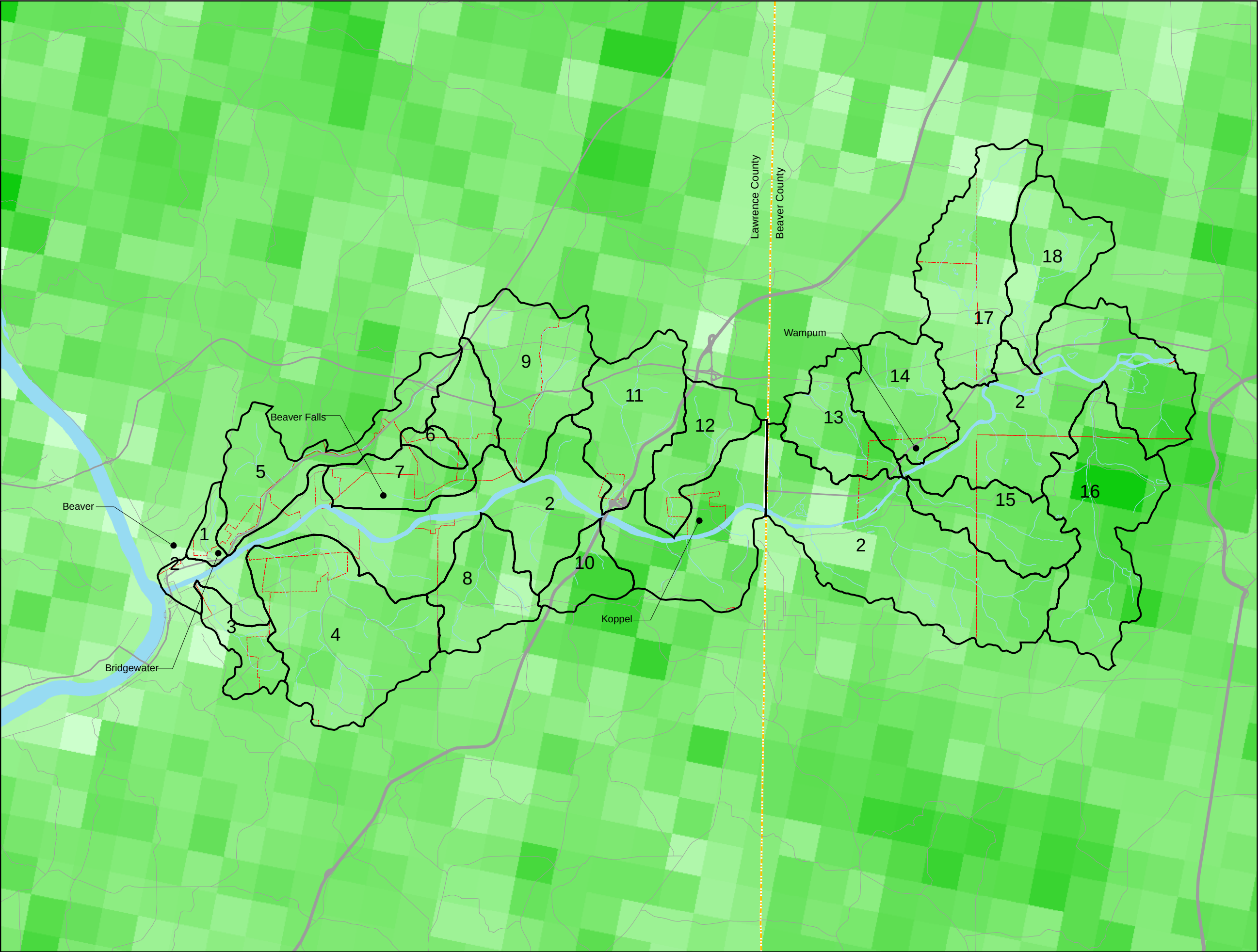
Threats to the vegetation include mineral extraction, development, utility and road right-of-ways, certain forestry practices, habitat fragmentation, and gypsy moths.

As identified by the Commonwealth of Pennsylvania agricultural security areas are:

- Noncontiguous farm parcels that are at least 10 acres in size. The farm tracts needed to create a new 250 acre or larger agricultural security area do not have to be under the same ownership or even be located in the same municipality. The Agricultural Area Security Law (Act 43 of 1981) allows for the creation of joint municipality agricultural security areas.
- The property should be viable agricultural land. Cropland, pasture, and woodland can all be included in an agricultural security area.
- At least 50% of the land should be in Soil Capability Classes I-IV, as defined by the county soil survey.
- The property must be zoned to permit agricultural uses.

Agricultural security areas are created by local municipalities in cooperation with individual landowners who agree to collectively place at least 250 acres in an agricultural security area. Another form of land classification within the Southwestern Pennsylvania region is agriculture preservation. These lands are typically recorded locally and/or on the County basis.¹⁹

¹⁹ <http://www.agriculture.state.pa.us/agriculture/cwp/view.asp?a=3&q=129076>



Legend

- Watershed Boundary
- County Boundary
- Rivers/Streams/Lakes
- Major Highways
- Major Roads
- Minor Roads

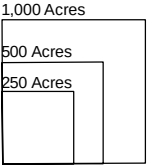
Forest Density¹

Most Significant Forest Density

Least Significant Forest Density

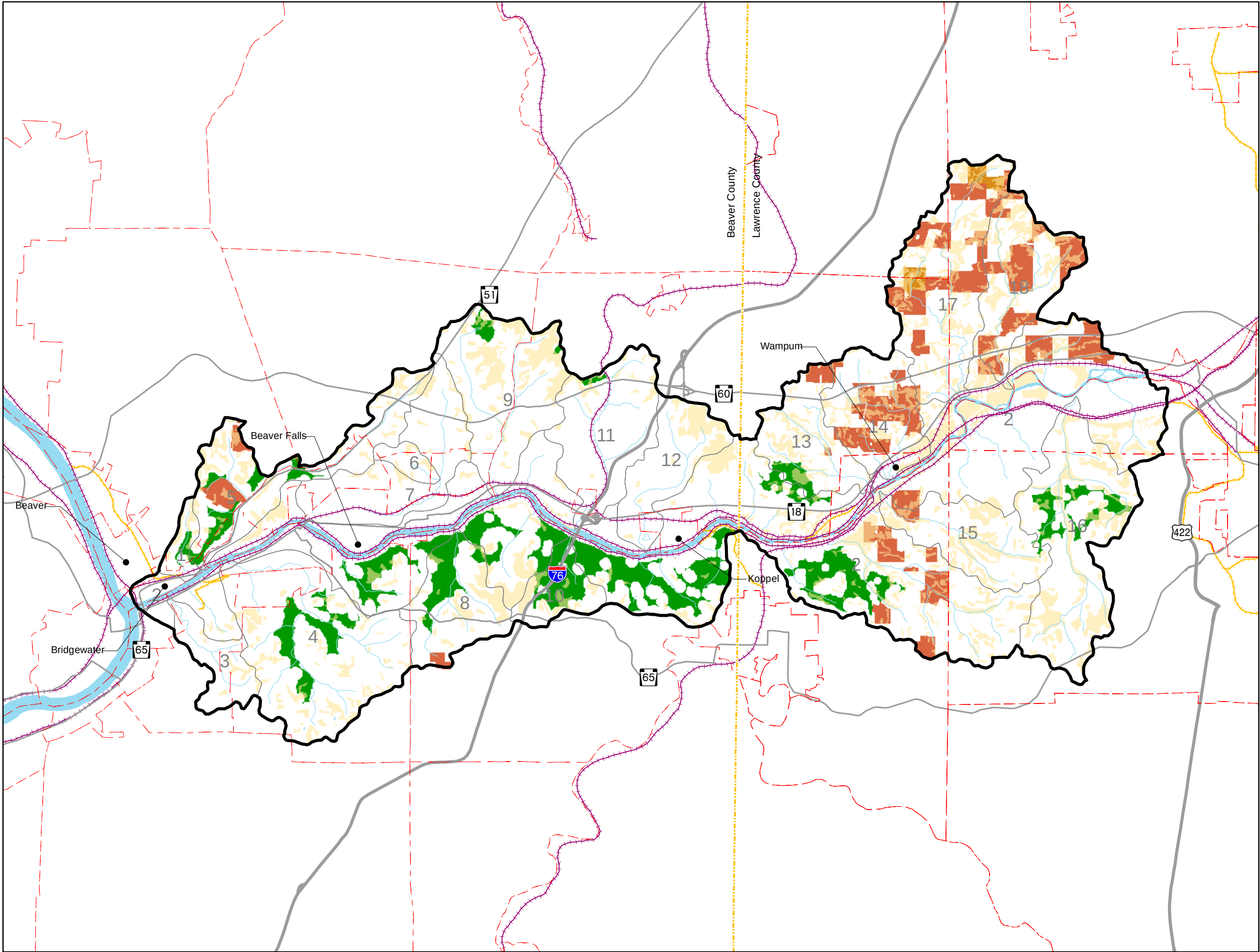
1. Represents trees per acre.

- Subwatersheds**
- 1. Hamilton Run
 - 2. Beaver River
 - 3. McKinley Run
 - 4. Block House Run
 - 5. Brady Run
 - 6. Grimms Run
 - 7. Walnut Bottom Run
 - 8. Bennett Run
 - 9. Wallace Run
 - 10. Thompson Run
 - 11. Clarks Run
 - 12. Stockman Run
 - 13. Wampum Run
 - 14. Eckles Run
 - 15. Snake Run
 - 16. McKee Run
 - 17. Jenkins Run
 - 18. Edwards Run



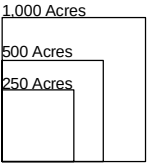
**FOREST DENSITY
BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC



- Legend**
- Study Area Boundary
 - Watershed Boundary
 - County Boundary
 - Municipal Boundaries
 - Highways
 - Major Roads
 - Rivers/Streams/Lakes
 - Prime Agriculture Soil
 - Agricultural Security Area
 - Agricultural Preservation Area
 - Habitable Woodland Area

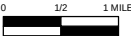
- Subwatersheds**
1. Hamilton Run
 2. Beaver River
 3. McKinley Run
 4. Block House Run
 5. Brady Run
 6. Grimms Run
 7. Walnut Bottom Run
 8. Bennett Run
 9. Wallace Run
 10. Thompson Run
 11. Clarks Run
 12. Stockman Run
 13. Wampum Run
 14. Eckles Run
 15. Snake Run
 16. McKee Run
 17. Jenkins Run
 18. Edwards Run



**WOODLAND AND
AGRICULTURE AREAS
BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.06.28



Invasive Species

Invasive species are species that grow aggressively, spread, and displace other species. They are difficult and expensive to control and can dominate whole areas, thereby threatening native plant and animal species. Most invasive species arrive from overseas; however, any that have been introduced into and thrive in an area where they were not found before (e.g. from another geographic region) is an invasive species.²⁰ Invasive species are some of the most precarious and unnoticed forms of environmental decline. Once established, seeds of invasives can be spread by water, wind, or animal droppings, further exacerbating the problem. Invasive species pose a threat to the biodiversity of the native flora and fauna along the Beaver River as well as across the United States. They easily overtake areas, especially where there is currently little vegetation. Once established, they grow rapidly, overtaking, for example, riverbanks, pastures, and public areas like parks and trails. They can crowd out native species that cannot survive the competition for nutrients and sunlight, thus leading to the decline of local biodiversity, and the increase in invasive monocultures.

Invasive plant species that occur in the study area include Japanese knotweed (*Polygonum cuspidatum*), multiflora rose (*Rosa multiflora*), garlic mustard (*Alliaria petiolata*), Japanese barberry (*Berberis thunbergii*), purple loosestrife (*Lythrum salicaria*), and Amur honeysuckle (*Lonicera maackii*). These species tend to invade areas that have been disturbed, or stripped of native vegetation, which allows for easy propagation of invasives.

There is no comprehensive database of invasive species sites in the study area or by county. However, County Natural Heritage Inventories, discussed later in this chapter, note when invasive species threaten specific sites.



Purple Loosestrife



Japanese Knotweed

²⁰ Adapted from “Invasive Plants in PA,” a PA DCNR brochure.



Multiflora Rosa



Garlic Mustard



Japanese Barberry



Amur Honeysuckle

Invasive plants are difficult to eradicate, but measures can be taken to control their spread. For example, Japanese knotweed spreads by rhizomes, or shallow underground stems; therefore, knotweed should not be removed by digging because rhizomes in the soil can wash downstream and invade other shorelines. Instead, knotweed can be controlled by cutting it four times a year, by planting native trees to shade it (knotweed is shade intolerant), or by spraying the plant with an herbicide that is safe to use near water.

Multiflora rose is currently being controlled by a viral pathogen disease called 'rose-rosette.'

Animals can be invasive species as well. The zebra mussel (*Dreissena polymorpha*) has been identified in the Beaver River watershed. This species is a major problem in other watersheds and has dispersed rapidly throughout the Great Lakes and major river systems. This is accomplished through 'hitch-hiking' on boats navigating these watercourses. Its rapid expansion into connected and unconnected waterways was probably due to barge traffic and recreational boating where it is theorized that attached mussels fell off during routine navigation. While they are not a substantial problem yet, zebra mussels can affect water treatment plants by clogging the water intake valves in the rivers. Additionally, it is noteworthy to mention that under cool, humid conditions, zebra mussels can stay alive for several days out of water; therefore boaters that utilize the river and other watercourses should take extra precautions to inspect their watercraft before and after boating, especially when transporting the watercraft to a different body of water.

Invasive pests in this region that are a threat to forested areas include the gypsy moth (*Lymantria dispar*) and the hemlock wooly adelgid (*Adelges tsugae*).



Zebra Mussel



Gypsy Moth



Hemlock Woolly Adelgid

Pennsylvania Natural Heritage Program (PNIP)

Formerly the Pennsylvania Natural Diversity Inventory (PNDI)

The PNHP is a partnership among the Pennsylvania Bureau of Forestry, Western Pennsylvania Conservancy, and The Nature Conservancy who “conduct inventories and collect data to identify and describe Pennsylvania’s rarest and most significant ecological features, which are needed for conservation, development planning, and natural resource management.”²¹ See Appendix 3 for a list of those species in the corridor.

Important Habitats

Features listed below can be found on the Sensitive Resources Inventory Map.

Natural Heritage Inventory

Areas within a Natural Heritage Inventory are classified as:

- Biological Diversity Area (BDA), which contains one or more locations of plants, animals, or natural communities recognized as a state or federal species of concern and/or a high quality example of areas supporting exceptional natural diversity;
- Landscape Conservation Area (LCA), defined as a large contiguous area that is important because of its size, open space, habitats, and/or the inclusion of one or more BDAs; or
- Managed Lands, which are owned or leased properties with importance, or potential importance, to the overall maintenance of and protection of the ecological resources of the county. They can be public or private.

Lawrence County’s NHI lists the following areas and recommendations in the Beaver RCMP study area.

²¹ <http://www.dcnr.state.pa.us>

Beaver River Island BDA

Description: Three islands and adjacent floodplains along the Beaver River below the confluence of the Shenango and Mahoning Rivers and one island upstream on the Shenango River. Islands and floodplains forested with a sycamore (river birch) box-elder floodplain forest. Area contains numerous wildflowers.

Threats: Forestry activity has occurred on the islands. Flooding cycles are important in maintaining the forest community. Exotic species (Japanese knotweed, multiflora rose) occur here.

Beaver River Floodplain BDA

Description: A large floodplain forest of mature sycamore-box-elder and numerous wildflowers.

Threats: Habitat relies on occasional flooding from Beaver River, which is controlled by dams on the Shenango and Mahoning Rivers; exotic species (multiflora rose, garlic mustard, dame's rocket); and loss of forest regeneration due to deer browsing.

Recommendations: Maintain natural flooding cycles. Allow floodplain forests to mature without timbering. Control exotic species. Control deer herds. Avoid canopy removal.

CS and M Mine BDA

Description: Habitat for two Pennsylvania animal species of concern. These species require cool temperatures, large spaces to hibernate, an area to forage outside of the cave, and minimal disturbance within immediate habitats.

Threats: Inadvertent blockage of the entrance; human visitation and/or vandalism; disturbance of airflow; removal of the canopy; additional mining; other activities near the mine.

Recommendations: Make landowner aware of the significance of the mine; gate the mine to limit access; evaluate activities near the mine to determine impacts to the mine and animals.

Rock Point BDA

Description: A contiguous forest at the confluence of Connoquenessing Creek and the Beaver River that is surrounded by non-forest. It provides a corridor for migrating birds and other animals moving from the Beaver River to the Slippery Rock watershed. Steep sandstone cliffs are above the Connoquenessing Creek. The Creek itself contains whitewater rapids. Numerous wildflowers occur in the area.

Threats: Invasive species (multiflora rose, garlic mustard, Japanese barberry, and amur honeysuckle).

Recommendations: Protect the unfragmented forest; eradicate invasive species.

Beaver County's NHI shows the following areas in the Beaver RCMP study area.

Ohio River BDA

Description: The river is a designated BDA because it serves as habitat for several fish species of special concern. These fish have returned to the area because of water quality improvements, but there is not enough information about the extent of their range or whether they are successfully inhabiting the river.

Threats: Water pollution

Recommendations: Continued and more stringent restrictions and regulations placed on industrial, residential, and commercial development along the river and its tributaries; careful monitoring and enforcement of regulations of all activities.

Beaver River Confluence Slopes BDA

Description: A habitat south of the Connoquenessing Creek's confluence with the Beaver River that supports a plant species of special concern. It is located near abandoned and active railroad tracks.

Threats: Easy access and the use of ATVs and dirt bikes.

Recommendations: Discuss protection of site with landowners; limit activity to passive recreation.

Beaver River Island BDA

Description: A small island – the only island habitat in the Beaver County section of the Beaver River – and the river and floodplains associated with the island. A Floodplain Forest Community exists here. Also provides a resting space for bird species migrating along the river.

Threats: Potential dredging; docks and campgrounds.

Recommendations: Avoid use of island until natural qualities have recovered; develop an impact study of potential recreational uses.

Big Beaver Boro Community Park

Description: A 90 acre park that is mostly forested slopes along a tributary to Clarks Run.

Threats: none listed

Recommendations: Allow forested areas to recover from past disturbances; avoid development or any other activity that would impact natural qualities.

Brady Run LCA

Description: Encompasses two large BDAs (along the North and South Branches of Bradys Run) and the Brady Run County Park.

Threats:

Recommendations: County should acquire private lands adjacent to the park property in areas that have been identified as needing added protection.

Brady Run County Park

Description: A 1,465 acre county park that contains wooded slopes, uplands, and stream bottoms. Managed for recreation, therefore the park has development throughout for recreational purposes.

Threats:

Recommendations: Buffer land around park by purchasing property or conservation easements.

South Branch Valley BDA

Description: A community/ecosystem conservation area that represents some of the better county examples of a Mesic Central Forest Community.

Threats: ATVs and mountain bikes; upland development, which impacts the hydrology of surrounding slopes.

Recommendations: Restrict mountain bikes and motorized vehicles on hiking trails; educate landowners and developers about ecologically sensitive approaches to farming and development.

North Branch Valley BDA

Description: Stream valley between Routes 60 and 51 of fairly contiguous forested slopes. It is a special species habitat, high diversity area, and community/ecosystem conservation area

Threats: Expansion or construction of additional parking lots and hiking trails; use of ATVs and other motorized vehicles.

Recommendations: Limit activities where the species of special concern occurs; limit activities that would further fragment natural communities.

New Brighton Valley BDA

Description: A large river tributary that remains in a fairly natural condition along the Beaver River. A steep-sided ravine and series of waterfalls also characterize the river valley.

Threats: Residential development in the uplands; construction of railroad tracks.

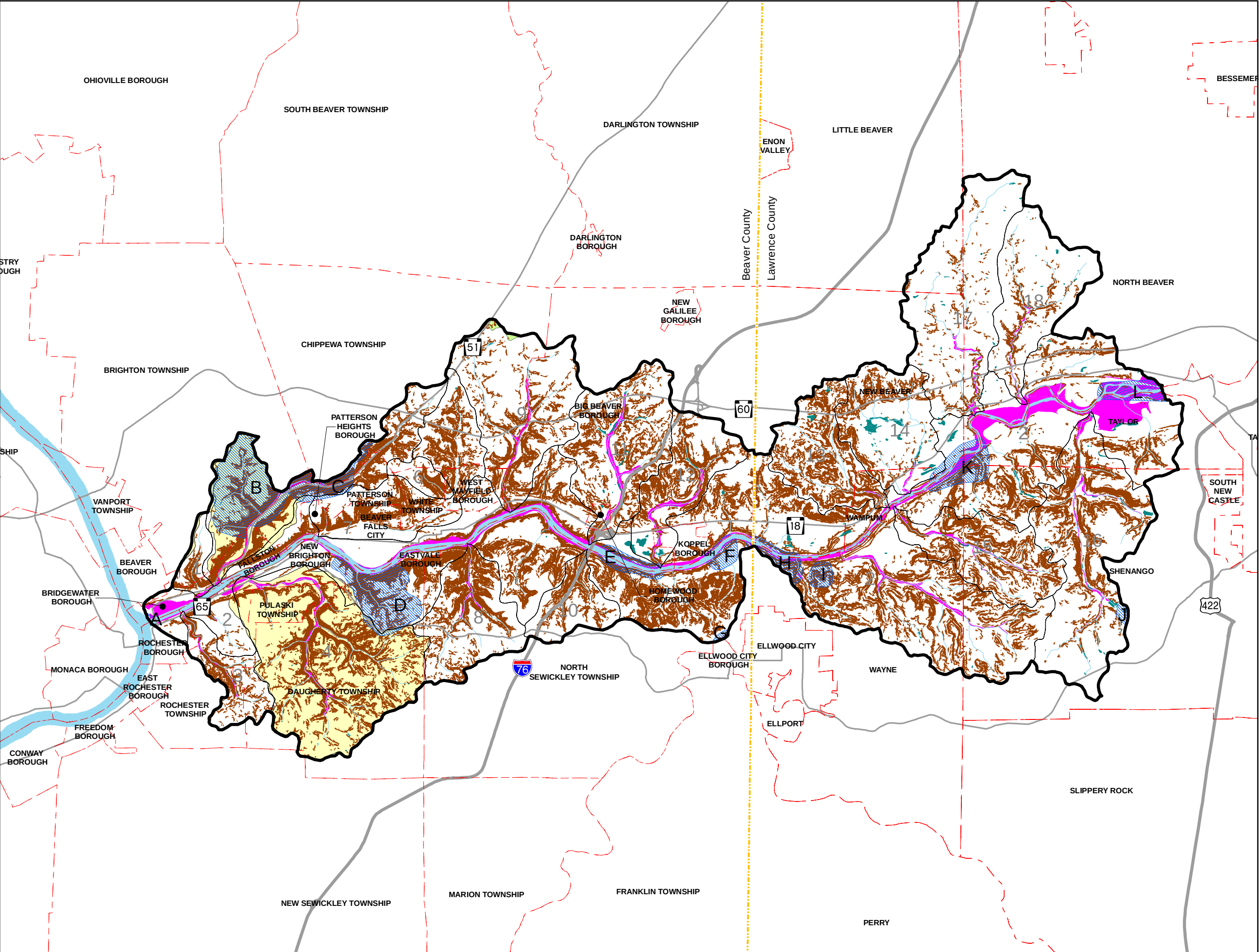
Recommendations: Allow upland forests to remain intact and undisturbed; minimize disturbance at railroad tracks; maintain a buffer zone for this site.

Homewood Falls

Description: 30 foot high waterfall along Clarks Run

Threats: none listed

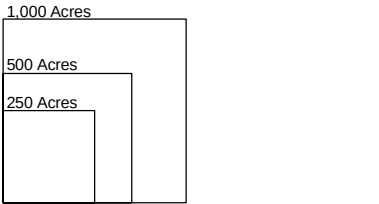
Recommendations: Develop area into park



- Legend**
- Study Area Boundary
 - Watershed Boundary
 - County Boundary
 - Municipal Boundaries
 - Highways
 - Major Roads
 - Rivers/Streams/Lakes
 - Biological Diversity Areas
 - >25% Slopes
 - Flood Prone Areas
 - Prime Warm Water Fish Habitat Watershed
 - Good Warm Water Fish Habitat Watershed
 - Marsh Related (Palustrine) Wetlands
 - River Related (Riverine) Wetlands

- Biological Diversity Areas**
- A. Ohio River BDA
 - B. South Branch Valley BDA
 - C. North Branch Valley BDA
 - D. New Brighton Valley BDA
 - E. Beaver River Island BDA
 - F. Beaver River Confluence Slope BDA
 - G. Steifel Park Mine BDA
 - H. Rock Point BDA
 - I. CS & M Mine BDA
 - J. Gardner Swamp BDA
 - K. Beaver River Flood Plain BDA
 - L. Beaver River Islands BDA

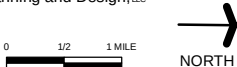
- Sub-Watersheds**
- 1. Hamilton Run
 - 2. Beaver River
 - 3. McKinley Run
 - 4. Block House Run
 - 5. Brady Run
 - 6. Grimms Run
 - 7. Walnut Bottom Run
 - 8. Bennett Run
 - 9. Wallace Run
 - 10. Thompson Run
 - 11. Clarks Run
 - 12. Stockman Run
 - 13. Wampum Run
 - 14. Eckles Run
 - 15. Snake Run
 - 16. McKee Run
 - 17. Jenkins Run
 - 18. Edwards Run



**SENSITIVE RESOURCES
INVENTORY
BEAVER RIVER CONSERVATION
AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: July 2006
1989.05.21r4



Wetlands

Wetlands, commonly known as marshes, bogs, swamps, or shallow ponds, are areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support a prevalence of vegetation typically adapted for life in saturated soil conditions.²²

Wetlands are important ecological resources; they filter runoff (and thus help to purify water quality), absorb heavy water flow (which alleviates flooding), provide habitat for many species, and promote recreation and tourism.

Most wetland location data is obtained from on-the-ground surveys, such as a County Natural Heritage Inventory (NHI). The NHI has limitations in data gathering, so it should not be considered an exhaustive list for wetlands in those counties.

Steep Sloped Areas and Landform

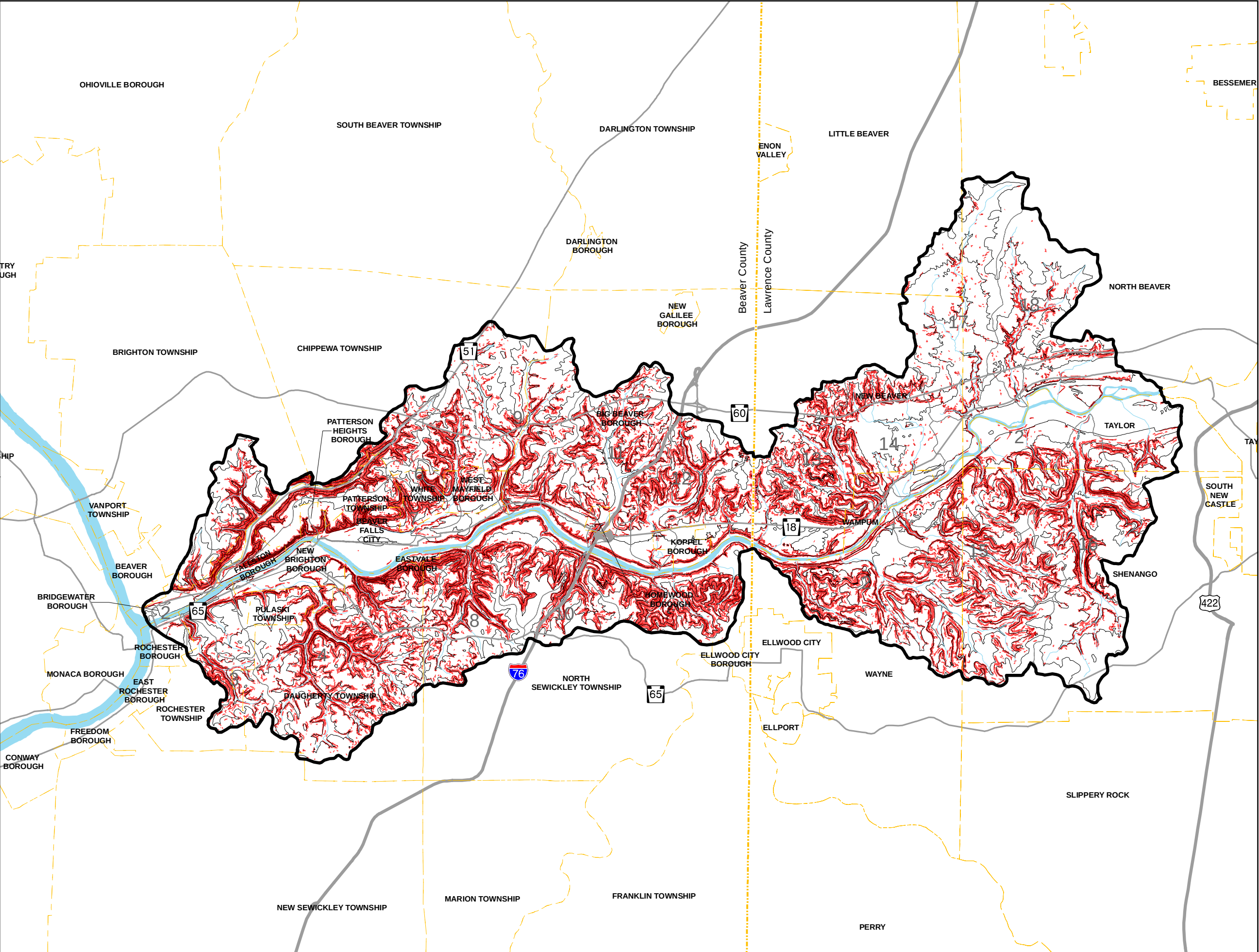
Steep forested slopes provide habitat for terrestrial species within the plan area largely due to the fact that these areas are unlikely to be developed. These areas are typically composed of pole stage and mature tree species, which contain diverse under-stratum habitats.

The Steep Slope Analysis Map identifies the locations of steep slopes. Slope grading is the percent (rise over run) expressing the steepness of a hill where zero indicates level, with respect to gradient and increasing numbers correlate to more vertical inclinations.

Slope refers the steepness of land and is usually defined as a percentage where zero percent slope indicates flat, level ground and increasing percentage represent greater vertical change. As an example of this calculation, a surface that rises 2 vertical feet over 10 feet of horizontal distance equals a slope of 20 percent. Within the Beaver River Corridor Study Area, land is generally most steep along the Banks of Beaver River and the streams that feed it. Slopes that are greater than 25 percent are typically considered “steep” and generally not suitable for intense development. Slopes between 25 and 40 percent gradient comprise approximately 12 percent of the study area. Approximately 6 percent of the Study Area’s land is characterized by slopes that are equal to or greater than 40 percent in gradient.

An analysis of the Study Area’s landform illustrates patterns of topography and elevation above sea level. Landform is shaped by geological forces and natural erosion processes caused by wind and water. The lowest point of the Study Area is the confluence of the Beaver and Ohio Rivers.

²² DEP Wetlands Factsheet www.dep.state.pa.us



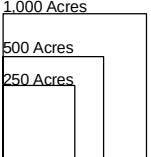
Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundaries
- Rivers/Streams/Lakes
- Highways
- Major Roads
- 100' Topographic Contour

Steep Slopes

>25-40%	7,439 acres
>40%	3,580 acres

- Sub-Watersheds**
1. Hamilton Run
 2. Beaver River
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 18. Edwards Run



Riparian (Streamside) Corridors

Riparian zones refer to the area between the land and water along a stream, river, lake, pond, or wetland. In this region of the country, riparian buffers are typically forested and are important to the ecological health of the waterways as they stabilize the stream bank, reduce erosion, decrease nutrient loads from runoff, provide habitat, maintain water temperature, and provide a source of food for aquatic life.

Riparian buffers are vegetated areas adjacent to a watercourse that provide bank stabilization as well as aquatic and wildlife habitat protection. Riparian buffers have been shown to be effective in controlling non-point source pollution by removing nutrients, especially nitrogen and sediment. The majority of the study area riparian buffers are classified as Forested Areas. See Riparian Buffer Analysis Map.

A wealth of information can be found on-line about the importance of riparian buffers. To start, visit the PA DEP's website (www.dep.state.pa.us) and learn about the Pennsylvania Stream ReLeaf Program. Factsheets and a riparian buffer toolkit can be downloaded.